The Quantum Composition Paradox

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Quantum theory does not generally permit the probability laws obtained from individual unitary steps by the Born rule to be sewn into a consistent genealogy; we classify the exceptions and show that faithful composition can hold from an initial boundary yet fail after an internal restart. For finite-dimensional, composition-closed unitary families, universal composition holds exactly for unitary monomials, the phase-dressed permutations whose Born kernels realize reversible deterministic state machines. Prescribed sequences evade this obstruction. We classify all pairs of qubit unitary steps, give a necessary-and-sufficient criterion for pairs of qutrit unitary steps, and prove that a boundary-stable unitary sequence on a d-dimensional Hilbert space contains at most d fully mixing steps, with equality in every prime dimension. We also construct arbitrarily long genuinely mixing sequences that compose from their initial boundary but fail after an internal restart, and a qutrit-controlled two-qubit realization with active interference. We define a Born–Chapman–Kolmogorov current that vanishes exactly when coherent and stepwise-checked endpoint laws agree, together with an associated measure of how many bits the endpoint reveals about the intermediate checking schedule. This state-machine connection provides a foundation for a quantum theory of music, in which unitary operations are notes and temporal boundaries are cues. Musical-transition prediction is proved PromiseBQP-complete, and the stepwise patterns that preserve a genealogy specify rules for rhythm, whereas the exceptional failure of that genealogy from an internal cue is the quantum music paradox.

Key words: Born rule, stochastic divisibility, quantum histories, quantum state machines, quantum process theory, quantum music, quantum circuits

1 The composition problem of quantum mechanics

Quantum theory offers two distinct routes to the same final time. If observations remain unchecked, amplitudes combine and interfere; if an intermediate observation creates a record, probabilities combine instead. The double slit makes the resulting disagreement vivid. Its experimental lineage runs from Young's interference experiment to Jönsson's electron realization [1, 2]; Born's probability rule and Bohr's complementarity supply its enduring conceptual vocabulary [3, 4].

Feynman eventually said that this phenomenon "*has in it the heart of quantum mechanics. In reality, it contains the only mystery*" [5]. Here that mystery appears as a failure of composition: coherent amplitudes compose, but their Born probabilities generally do not. Classifying when they do compose poses a quantum–classical interface challenge and raises a sharp question: does a faithful genealogy survive restriction to a newly prepared internal boundary, as it must for a consistent temporal model that can be restarted from that boundary?

The double slit provides the familiar negative case: stepwise and endpoint-checked routes assign different probabilities. This question narrows the measurement problem. Quantum mechanics has long supplied probability laws for successive observations [6, 7]; in the fixed-context setting used here, summing over intermediate outcomes gives the stepwise-checked Born probability kernel. We first classify when it equals the endpoint Born kernel of the same evolution without intermediate checks, and then whether that same equality survives an internal boundary.

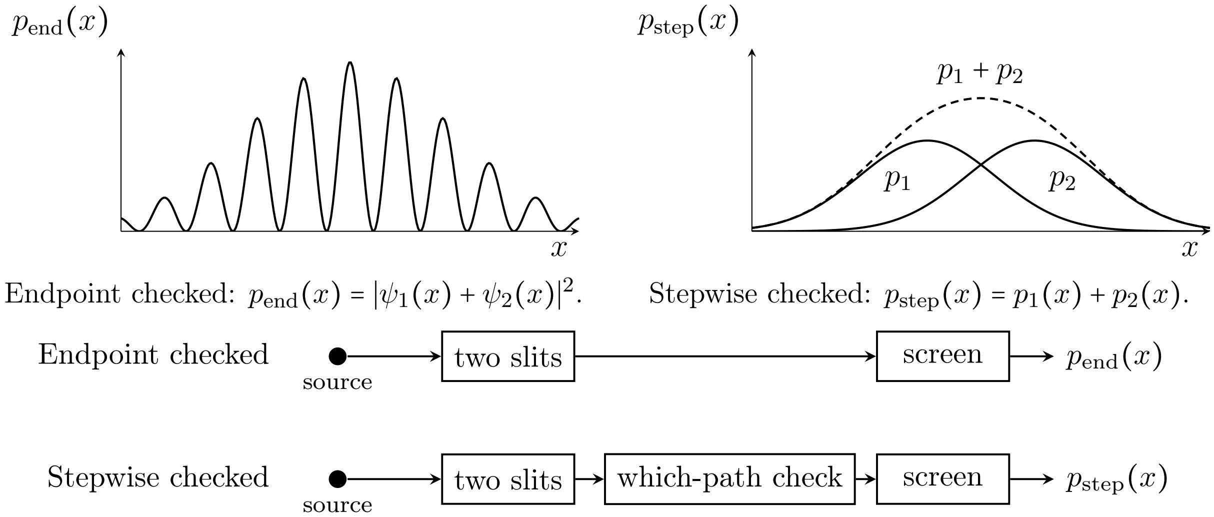


Figure 1: **The double slit as a quantum–classical composition failure.** Endpoint-checked alternatives add as amplitudes and interfere. Stepwise-checked alternatives add as probabilities. The two processes assign different distributions to the final outcomes.

Let $X = \{0, \dots, d-1\}$ label the basis of a d-dimensional Hilbert space $\mathsf{H} \cong \mathbb{C}^d$, and fix the atomic context $\mathcal{C} = \{P_x = |x\rangle\langle x| : x \in X\}$, with $\sum_{x\in X} P_x = \mathbb{1}$. A $\mathcal{C}$-diagonal initial (entrance) state has the form $\rho = \sum_{x\in X} q(x) P_x$ for any probability distribution q over X. For any unitary $U \in \mathrm{U}(d)$, write $\mathsf{B}(U) = \mathsf{B}_{\mathcal{C}}(U)$ for its *Born kernel*, the transition matrix in this context:

$$\begin{aligned} \mathsf{B}(U) &:= U \odot \overline{U}, \\ \langle y|\mathsf{B}(U)|x\rangle = \langle yy|\big(U \otimes \overline{U}\big)|xx\rangle &= |\langle y|U|x\rangle|^2. \end{aligned} \tag{1}$$

Here $\odot$ is elementwise (Hadamard) product and the overbar denotes entrywise complex conjugation in the fixed context. This matrix element is the transition probability from $|x\rangle$ to $|y\rangle$ under U. Each $\mathsf{B}(U)$ is unistochastic, hence doubly stochastic, and therefore a valid classical transition map over the chosen outcomes. Every kernel identity below holds for every context-basis preparation and hence for every probability distribution over that context. The context $\mathcal{C}$ fixes the basis in which the entrywise conjugation and Hadamard product are taken.

We use $\circ$ for chronological composition of both unitaries and their stochastic kernels, where $V \circ U$ means that U acts before V.

A family of Born kernels assigned to allowed contiguous passages is a *Born-induced stochastic genealogy* when, at every allowed cut, the kernel of the whole passage equals the chronological product of the kernels of its subpassages.

For a unitary U followed by a unitary V, define what we call the signed *Born–Chapman–Kolmogorov current*, or *Born-composition defect*, and its normalized Frobenius norm by

$$\mathcal{J}_{\mathcal{C}}(V,U) := \mathsf{B}_{\mathcal{C}}(V \circ U) - \mathsf{B}_{\mathcal{C}}(V) \circ \mathsf{B}_{\mathcal{C}}(U), \tag{2}$$

$$0 \le \nu_{\mathcal{C}}(V,U) := \frac{1}{\sqrt{d-1}} \|\mathcal{J}_{\mathcal{C}}(V,U)\|_{\mathrm{F}} \le 1. \tag{3}$$

Here $d \ge 2$, and the upper bound is sharp by Theorem D.4. This terminology denotes the signed departure from the Chapman–Kolmogorov equation which is a faithful consistency relation for classical transition kernels in stochastic processes and statistical mechanics [8, 9, 10].

To quantify the information revealed from an intermediate measurement, prepare a uniformly chosen context-basis state. In this two-step experiment, X is a uniformly sampled, recorded input basis label, S is an independent fair bit selecting whether to insert the unread intermediate context check, and Y is the recorded output label. Write $\mathcal{I}(S;Y \mid X)$ for the conditional mutual information about the checking schedule, with $0 \le \mathcal{I} \le 1$ bit per run in the uniform-input, two-step case of Proposition D.2. Here both recorded variables X and Y take values in the same outcome alphabet X associated with $\mathcal{C} = \{P_x : x \in X\}$.

Figure 2 gives equivalent formulations of the quantum–classical consistency law for a faithful Born-kernel genealogy. The endpoint probabilities coincide, whereas the intermediate check may still change the quantum state.

Quantum composition **Stochastic composition**

$$\begin{tikzcd}[column sep=small]
 & (U,V) \arrow[dl, "\text{compose}"'] \arrow[dr, "\text{evaluate}"] & \\
V \circ U \arrow[d, "\text{evaluate}"'] & & (V \odot \bar{V},\ U \odot \bar{U}) \arrow[d, "\text{compose}"] \\
(V \circ U) \odot (\bar{V} \circ \bar{U}) \arrow[rr, equal] & & (V \odot \bar{V}) \circ (U \odot \bar{U})
\end{tikzcd}$$

The diagram commutes $\Longleftrightarrow \mathsf{B}_{\mathcal{C}}(V \circ U) = \mathsf{B}_{\mathcal{C}}(V) \circ \mathsf{B}_{\mathcal{C}}(U)$

$\Longleftrightarrow \mathcal{J}_{\mathcal{C}}(V,U) = 0 \Longleftrightarrow \nu_{\mathcal{C}}(V,U) = 0 \Longleftrightarrow \mathcal{I} = 0.$

Figure 2: **The quantum–classical consistency law for faithful Born-kernel composition.** The left route composes the unitaries before Born evaluation; the right route composes their Born kernels.

Several related notions must be separated. Pollock et al. and Milz and Modi distinguish operational process-tensor Markovianity from weaker divisibility tests: agreement at the level of intermediate maps need not supply a full multitime Markov process [11, 12]. Multitime Kolmogorov consistency asks when inserting or removing measurements preserves the relevant statistics [13, 14] and [15, 16]. Our equality is the complete specialization to prescribed closed-system unitary passages. Barandes develops a formulation of quantum theory in terms of indivisible stochastic processes. His framework identifies permutations as stochastic maps with stochastic inverses and relates generic Born-kernel noncomposition to interference [17, 18, 19]. We classify the exact relevant cases. The Born-kernel composition law on a unitary semigroup yields unitary monomials, whereas prescribed genuinely mixing unitaries can compose on some passages and fail on others; composition need not survive all shifts inside the preparation boundary. The framework also supplies the Born–Chapman–Kolmogorov current and its exact schedule-information measure, a consistency theorem for reversible state machines, and an operation-supplied quantum counterpart of Xenakis-style stochastic music [20, 21, 22]. In the circuit score model, musical-transition prediction is proved to be PromiseBQP-complete.

The conditions under which the diagram in Fig. 2 commutes admit an equivalent path-sum formulation. For entrance x, exit y, and paths $\gamma = (x, z, y)$ through intermediate context outcomes z, write $\mathcal{A}(\gamma) = \langle y|V|z\rangle\langle z|U|x\rangle$. Then

$$\begin{aligned}\langle y|\mathcal{J}_{\mathcal{C}}(V,U)|x\rangle &= \Big|\sum_{\gamma} \mathcal{A}(\gamma)\Big|^2 - \sum_{\gamma} |\mathcal{A}(\gamma)|^2 \\ &= \sum_{\gamma \neq \gamma'} \mathcal{A}(\gamma)\overline{\mathcal{A}}(\gamma').\end{aligned} \tag{4}$$

Here $\gamma \neq \gamma'$ means $z \neq z'$; the overbar denotes complex conjugation.

The diagram commutes exactly when this aggregate interference vanishes for every entrance–exit pair; the normalized current $\nu_{\mathcal{C}}(V, U)$ quantifies the failure of composition.

Equation (4) makes the distinction from consistent histories explicit. Weak or medium decoherence for every context-basis entrance implies endpoint Born-kernel composition [23, 24, 25, 26]. However, the converse can fail: diagram commutation requires only aggregate interference cancellation, and Appendix C.7 gives a multistep passage that composes while violating even weak decoherence. The full multistep path-sum derivation is given in Appendix C.5.

For a proposed Born-kernel semantics of finite-state quantum control or a quantum generalization of Xenakis-style music [20, 21, 22], vanishing of the current is the interval-level test that the locally supplied kernels reproduce the endpoint distribution of the specified coherent passage. Boundary stability imposes this test on every required interval.

The argument begins with three results. Theorem 2.1 classifies universal Born-kernel composition, Theorem 4.2 isolates the boundary-sensitive separation, and Theorem E.3 gives a dense two-qubit realization. Section 5 develops the application to quantum music: composition identifies admissible rhythms, and musical-transition prediction is proved PromiseBQP-complete in the circuit-universal score model (Definition 5.1 and Theorem 5.2).

2 Universal Born-kernel composition

The standard formulation of quantum theory treats reversible transformations as forming the unitary group $\mathrm{U}(d)$. Thus every U comes with an inverse $U^\dagger$, with $U^\dagger \circ U = U \circ U^\dagger = \mathbb{1}$. If the Born images are also to form a genealogy under the same composition law, then

$$\mathsf{B}(U^\dagger \circ U) = \mathsf{B}(\mathbb{1}) = \mathsf{B}(U)^\top \circ \mathsf{B}(U). \tag{5}$$

Equation (5) holds if and only if U is a unitary monomial. We extend this result to universal composition. Fix a basis of distinguishable readout events, and let $\Gamma \subseteq \mathrm{U}(d)$ be a unitary semigroup of allowed transformations, meaning that

$$U, V \in \Gamma \implies V \circ U \in \Gamma.$$

No closure under adjoints is assumed, as Γ may or may not contain $U^\dagger$ when it contains U. The following theorem classifies universal composition.

Theorem 2.1 (Classification of Born-consistent unitary semigroups). *The identity*

$$\mathsf{B}(V \circ U) = \mathsf{B}(V) \circ \mathsf{B}(U) \quad \textit{for every } U, V \in \Gamma \tag{6}$$

holds if and only if every element of Γ is a unitary monomial in the context basis.

The proof is given in Appendix B.5. Every such map has the form $\Delta \circ \Pi$, where

$$\Delta = \operatorname{diag}\left(e^{-\imath\theta_1}, e^{-\imath\theta_2}, \ldots, e^{-\imath\theta_d}\right), \qquad \theta_1, \ldots, \theta_d \in \mathbb{R}, \tag{7}$$

is diagonal unitary and Π is a permutation matrix. The diagonal phases are gauge symmetries of the Born-kernel map:

$$\mathsf{B}_{\mathcal{C}}(\Delta) = \mathbb{1}, \qquad \mathsf{B}_{\mathcal{C}}(\Delta \circ \Pi) = \mathsf{B}_{\mathcal{C}}(\Pi) = \Pi. \tag{8}$$

Birkhoff's theorem places this result in its convex setting: every doubly stochastic matrix is a convex combination of permutation matrices [27]. At the unitary level, the conclusion becomes sharper: exact Born-kernel composition collapses the underlying unitary family itself to unitary monomials.

The semigroup classification has an immediate finite-state interpretation: universal Born-kernel composition selects unitary monomials, whose Born images are reversible deterministic state machines. The context outcomes supply the machine's finite states, and an output map assigns observable symbols to those states. Choosing a musical output alphabet gives this same compositional structure a musical interpretation. A compatible family of their permutation kernels supplies the multitime genealogy; the full Moore-machine corollary is stated in Appendix B.6.

3 Composition conditions for prescribed sequences

The classification of Born-consistent unitary semigroups leaves a different problem open: which prescribed pairs U, V satisfy the composition identity, without requiring it for every concatenation of those operations? More generally, which prescribed forward sequences give the same final outcome probabilities with and without the intermediate context checks?

For a prescribed sequence $U_1, \ldots, U_k$, the test becomes the boundary matrix identity

$$\mathsf{B}_{\mathcal{C}}(U_k \circ \cdots \circ U_1) = \mathsf{B}_{\mathcal{C}}(U_k) \circ \cdots \circ \mathsf{B}_{\mathcal{C}}(U_1). \tag{9}$$

It compares endpoint statistics of a coherent passage with the statistics obtained by recording the same fixed context at its intermediate boundaries. Taking the record generally changes the quantum state; the question is whether it changes the final distribution. The path-sum identity, its aggregate-interference criterion, and the formal separation of this theory from consistent histories are given in Appendices C.5 and C.7.

We say a Born kernel has *full support* when every matrix entry is strictly positive. A full-support kernel is called *genuinely mixing*. We give a genuinely mixing witness for which this equality holds and then show that it can hold from one initial boundary and fail as soon as the same steps are restarted from a later one.

A genuinely mixing single-qubit Born-kernel genealogy. Fix the computational context of one qubit and let $\sigma_x, \sigma_y, \sigma_z$ be the Pauli operators. The current in Eq. (2) measures the failure of a pair of unitary steps to compose over the two context outcomes. Every column sums to zero: the current moves probability but does not create it. A consistent genealogy requires it to vanish at every required cut.

Write

$$R_\sigma(\theta) := e^{-\imath\theta\sigma/2}. \tag{10}$$

Here $\sigma \in \{\sigma_x, \sigma_y, \sigma_z\}$, and every qubit Born kernel takes the form

$$K(\lambda) := \frac{1}{2}\begin{pmatrix} 1+\lambda & 1-\lambda \\ 1-\lambda & 1+\lambda \end{pmatrix}, \qquad -1 \leq \lambda \leq 1, \tag{11}$$

Thus a qubit step is genuinely mixing exactly when $-1 < \lambda < 1$. These stochastic maps obey

$$K(\lambda) \circ K(\mu) = K(\lambda \cdot \mu). \tag{12}$$

For $\mathsf{B}(U) = K(\mu)$ and $\mathsf{B}(V) = K(\lambda)$, this product equals $\mathsf{B}(V \circ U)$ if and only if the diagram in Fig. 2 commutes, equivalently if and only if $\nu_{\mathcal{C}}(V, U) = 0$.

For the cross-axis family $U = R_{\sigma_y}(\theta)$ and $V = R_{\sigma_x}(\phi)$,

$$\mathsf{B}(V \circ U) = K(\cos\phi \cdot \cos\theta) = \mathsf{B}(V) \circ \mathsf{B}(U)$$

for every θ, ϕ. Thus $\nu_{\mathcal{C}}(R_{\sigma_x}(\phi), R_{\sigma_y}(\theta)) = 0$ throughout the full two-parameter family, with θ and ϕ independent.

We now choose from this family a genuinely mixing pair of single-qubit unitary steps with distinct individual Born kernels. Put

$$\begin{aligned} U &= R_{\sigma_y}\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}\mathbb{1} - \frac{\imath}{2}\sigma_y, \\ V &= R_{\sigma_x}(\phi) = \sqrt{\frac{2}{3}}\mathbb{1} - \frac{\imath}{\sqrt{3}}\sigma_x, \qquad \cos\phi = \frac{1}{3}. \end{aligned} \tag{13}$$

The Born kernels are

$$\begin{aligned} \mathsf{B}(U) &= K(1/2) = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}, \\ \mathsf{B}(V) &= K(1/3) = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}. \end{aligned} \tag{14}$$

Nevertheless, the coherent two-step passage obeys their stochastic composition law:

$$\begin{aligned} \mathsf{B}(V \circ U) &= \mathsf{B}(V) \circ \mathsf{B}(U) \\ &= K(1/6) = \begin{pmatrix} \frac{7}{12} & \frac{5}{12} \\ \frac{5}{12} & \frac{7}{12} \end{pmatrix}. \end{aligned} \tag{15}$$

The computational context is the spectral context of σ_z, this witness satisfies $[U, \sigma_z] \neq 0$, $[V, \sigma_z] \neq 0$, and $[U, V] \neq 0$, even though $\nu_{\mathcal{C}}(V, U) = 0$. For each endpoint, the two contributing path amplitudes are both nonzero but differ in phase by $\pm\pi/2$, causing the conjugate cross terms in the Born kernel to cancel exactly. The equality therefore reflects an exact quantum cancellation rather than a monomial or one-path special case; Appendix C gives the calculation and the continuous $R_{\sigma_y}(\theta)$–$R_{\sigma_x}(\phi)$ family.

Beginning in $|0\rangle$, let $P_x = |x\rangle\langle x|$ and let the unread computational-basis projective Lüders channel be $\mathcal{M}_{\mathcal{C}}(\rho) = \sum_{x=0}^{1} P_x \rho P_x$; its outcome is discarded. Figure 3 lines up the two routes at their common boundaries.

The intermediate check is a non-selective rank-one projective Lüders measurement in

Endpoint-checked passage

$$\begin{pmatrix}1&0\\0&0\end{pmatrix}\xrightarrow{U}\begin{pmatrix}\frac34&\frac{\sqrt3}{4}\\\frac{\sqrt3}{4}&\frac14\end{pmatrix}\xrightarrow{\qquad\qquad V\qquad\qquad}\begin{pmatrix}\frac{7}{12}&\frac{\sqrt3}{4}+\frac{\imath\sqrt2}{6}\\\frac{\sqrt3}{4}-\frac{\imath\sqrt2}{6}&\frac{5}{12}\end{pmatrix}\xrightarrow{\mathcal{M}_{\mathcal{C}}}\begin{pmatrix}\frac{7}{12}&0\\0&\frac{5}{12}\end{pmatrix}$$

Stepwise-checked passage

$$\begin{pmatrix}1&0\\0&0\end{pmatrix}\xrightarrow{U}\begin{pmatrix}\frac34&\frac{\sqrt3}{4}\\\frac{\sqrt3}{4}&\frac14\end{pmatrix}\xrightarrow{\mathcal{M}_{\mathcal{C}}}\begin{pmatrix}\frac34&0\\0&\frac14\end{pmatrix}\xrightarrow{\qquad V\qquad}\underbrace{\begin{pmatrix}\frac{7}{12}&\frac{\imath\sqrt2}{6}\\-\frac{\imath\sqrt2}{6}&\frac{5}{12}\end{pmatrix}}_{\frac34 VP_0V^\dagger+\frac14 VP_1V^\dagger}\xrightarrow{\qquad\mathcal{M}_{\mathcal{C}}\qquad}\begin{pmatrix}\frac{7}{12}&0\\0&\frac{5}{12}\end{pmatrix}$$

$$\Pr(\text{end}=|0\rangle)=\frac34\cdot\frac23+\frac14\cdot\frac13=\frac12+\frac{1}{12}=\frac{7}{12}\qquad(|0\rangle\to|0\rangle,\ |1\rangle\to|0\rangle),$$

$$\Pr(\text{end}=|1\rangle)=\frac34\cdot\frac13+\frac14\cdot\frac23=\frac14+\frac16=\frac{5}{12}\qquad(|0\rangle\to|1\rangle,\ |1\rangle\to|1\rangle).$$

Figure 3: **Endpoint agreement for a pair of single-qubit unitary steps (U, V).** The unread intermediate projective check changes the quantum state while preserving the final context probabilities. The two recorded branches ending in $|0\rangle$ contribute 1/2 and 1/12; those ending in $|1\rangle$ contribute 1/4 and 1/6. For these unequal local kernels, the two endpoint distributions are both $(7/12, 5/12)$.

the computational (σ_z) context, equivalently context-basis dephasing [6]. Its outcome is discarded and the same qubit then evolves under V. It changes the state from pure to mixed, with the coherent and stepwise states differing by $\sqrt{3}\sigma_x/4$, yet leaves the endpoint marginal $(7/12, 5/12)$ unchanged. Thus two genuinely mixing prescribed steps with distinct Born kernels carry a genealogy despite this projective measurement disturbance. Appendices C and C.3 classify all pairs of qubit unitary steps and all pairs of qutrit unitary steps with this property; Appendix C derives the qubit witness from the general criterion.

Let R denote the final qubit before the context measurement. For the recorded entrance label X and independent fair checking bit S, write $I_c := I(S;Y \mid X)$ and $I_q := I(S;R \mid X)$. The two protocols produce different density operators for R, but identical context probabilities:

$$0 = I_c \le I_q \simeq 0.2504 \text{ bits}.$$

Thus the quantum state retains information about the intermediate check that the prescribed measurement conceals. Appendix D.1 gives the bound and calculation.

4 Composition at the quantum–classical divide

We provide a framework and prove that faithful composition from the initial boundary need not survive temporal restriction and fresh preparation at every internal boundary.

Let $U_1, \ldots, U_L$ be a prescribed sequence of qubit unitaries. For $1 \le a \le b \le L$, write

$$U_{b:a} := U_b \circ U_{b-1} \circ \cdots \circ U_a. \tag{16}$$

for the contiguous passage containing precisely the steps $U_a, \ldots, U_b$, with both endpoint steps included; in particular, $U_{k:k} = U_k$. Number the temporal boundaries $0, \ldots, L$, so U_k carries boundary $k-1$ to boundary k; consequently, $U_{b:a}$ begins at boundary $a-1$ and ends at boundary b. The boundary before U_1, boundary 0, is the *initial boundary*. Beginning at the boundary immediately before U_a, namely boundary $a-1$, means freshly preparing a $\mathcal{C}$-diagonal entrance state there and applying a contiguous passage $U_{b:a}$; this is an internal boundary when $2 \le a \le L$, and taking $b = L$ gives the full prescribed suffix. It is a new preparation boundary, not a pause and resumption of the earlier quantum state. There are two distinct consistency requirements.

Definition 4.1 (Prefix consistency). A sequence is *prefix-consistent* if

$$\mathsf{B}(U_{k:1}) = \mathsf{B}(U_k) \circ \cdots \circ \mathsf{B}(U_1) \quad \text{for every } 2 \le k \le L. \tag{17}$$

It is *boundary-stable*, equivalently subinterval-consistent, if

$$\begin{aligned} \mathsf{B}(U_{b:a}) = \mathsf{B}(U_b) \circ \cdots \circ \mathsf{B}(U_a),& \\ \text{for every } 1 \le a \le b \le L.& \end{aligned} \tag{18}$$

Here "*subinterval*" means an unbroken chronological passage $U_{b:a}$. A selection that skips an intermediate prescribed step, such as $U_3 \circ U_1$, represents a different circuit, not an additional subinterval condition.

Prefix consistency is the $a = 1$ case of boundary stability: it compares each coherent prefix with its fully stepwise-checked realization from the original boundary. Boundary stability makes the fixed-context sequential statistics Kolmogorov consistent under every insertion or deletion of an intermediate context measurement, for every context-diagonal entrance state; see Proposition C.14.

Theorem 4.2 (Qubit boundary separation). *For a fixed qubit context:*

1. *prefix-consistent sequences of arbitrary finite length exist with every step genuinely mixing;*

2. *consistency may fail on certain contiguous subintervals.*

Here the question is compatibility across preparation boundaries: even when every step is genuinely mixing, prefix consistency does not imply boundary stability.

To isolate the distinct issue of temporal-boundary instability, now consider the deliberately restricted *constant-amplitude infinite one-qubit sequence*

$$U_1 = R_{\sigma_y}\left(\frac{\pi}{3}\right), \qquad U_k = R_{\sigma_z}(\alpha_k) \circ R_{\sigma_x}\left(\frac{\pi}{3}\right), \qquad k \ge 2, \tag{19}$$

where $0 < \alpha_k < \pi/2$ is fixed by

$$\tan \alpha_k = \frac{\sqrt{3}}{2\sqrt{4^{k-1} - 1}}. \tag{20}$$

For its first three steps, $\alpha_2 = \arctan(1/2)$ and $\alpha_3 = \arctan(1/(2\sqrt{5}))$. Its third prefix composes, but the unchanged last two steps fail after an internal restart:

$$\begin{aligned} \mathsf{B}(U_3 \circ U_2 \circ U_1) &= K(1/8) = \mathsf{B}(U_3) \circ \mathsf{B}(U_2) \circ \mathsf{B}(U_1), \\ \mathsf{B}(U_3 \circ U_2) &= K\left(\frac{1 - 6/\sqrt{5}}{4}\right) \neq K(1/4) = \mathsf{B}(U_3) \circ \mathsf{B}(U_2). \end{aligned} \tag{21}$$

Thus $\nu_{\mathcal{C}}(U_3, U_2) \neq 0$. Every step has the same genuinely mixing Born kernel,

$$\mathsf{B}(U_k) = K(1/2) = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}, \tag{22}$$

The compensating phase changes with its position in the sequence. For the first ℓ-step passage $U_{\ell:1}$, one obtains

$$\mathsf{B}(U_{\ell:1}) = K(2^{-\ell}) = \mathsf{B}(U_\ell) \circ \cdots \circ \mathsf{B}(U_1) \qquad \text{for every } \ell \geq 1. \tag{23}$$

Thus the sequence is prefix-consistent at every finite length $\ell \geq 1$.

Yet restarting just before any U_r, $r \geq 2$, fails already over the next two steps:

$$\mathsf{B}(U_{r+1} \circ U_r) = K\left(\frac{1 - 3\cos\alpha_r}{4}\right) \neq K(1/4) = \mathsf{B}(U_{r+1}) \circ \mathsf{B}(U_r). \tag{24}$$

These steps retain the same individual Born kernels; only the temporal boundary has moved. Appendix C proves the recurrence and its sign-reversed companion, whose local kernel is $K(-1/2)$. It is not an inverse sequence.

For the first three steps, the one-step kernels and composition identities can be listed exhaustively. Since $\mathsf{B}(U_1) = K(1/2)$ is invertible, the

noncomposition of the suffix $U_3 \circ U_2$ persists under right composition with $\mathsf{B}(U_1)$.

Individual steps: $\mathsf{B}(U_1) = \mathsf{B}(U_2) = \mathsf{B}(U_3) = K(1/2)$.	
Composition holds	**Composition fails**
$\mathsf{B}(U_2 \circ U_1) = \mathsf{B}(U_2) \circ \mathsf{B}(U_1)$	$\mathsf{B}(U_3 \circ U_2) \neq \mathsf{B}(U_3) \circ \mathsf{B}(U_2)$
$\mathsf{B}(U_3 \circ U_2 \circ U_1) = \mathsf{B}(U_3) \circ \mathsf{B}(U_2 \circ U_1)$	$\mathsf{B}(U_3 \circ U_2 \circ U_1) \neq \mathsf{B}(U_3 \circ U_2) \circ \mathsf{B}(U_1)$
$\mathsf{B}(U_3 \circ U_2 \circ U_1) = \mathsf{B}(U_3) \circ \mathsf{B}(U_2) \circ \mathsf{B}(U_1)$	

A check after U_2 and checks after both U_1 and U_2 preserve the endpoint distribution, whereas a check only after U_1 changes it. Adding the second check restores an agreement that the first check alone destroys. With a recorded uniform entrance and a fair bit selecting whether to check the displayed cut, the two orphaned compositions give $(\mathcal{I}(S;Y \mid X), \nu_{\mathcal{C}}) \simeq (0.0834, 0.6708)$ and $(0.0204, 0.3354)$, respectively, with information in bits per run.

The general bounds $\mathcal{I}(S;Y \mid X) \leq 1$ bit and $\nu_{\mathcal{C}} \leq 1$ are proved in Appendices D.1 and D.2, respectively; the latter is sharp.

Nothing here requires a retrocausal interpretation. The endpoint law is determined by the forward-ordered unitary evolution of the entire prescribed passage, and no later operation alters an earlier recorded event. The issue is a failure of faithful temporal composition: locally induced Born kernels need not reproduce the coherent endpoint law after a fresh preparation at an internal boundary or when prescribed readouts are inserted or removed.

The qubit construction provides the minimal-dimensional boundary-sensitive witness. The qutrit analysis gives an exact condition for every prescribed pair of qutrit unitary steps.

Theorem 4.3 (Sharp full-support capacity). *Let $d \geq 2$ and let $U_1, \dots, U_L$ be unitary operations on $\mathbb{C}^d$. If the sequence is boundary-stable, then at most d one-step Born kernels $\mathsf{B}(U_k)$ have full support. Moreover, for every prime d, there exists a boundary-stable d-step sequence in which every one-step Born kernel has full support.*

The proofs are given in Appendices C.3 and D.4.

5 Quantum rhythm and transition prediction complexity

A musical composition combines a sequence of notes with its organization in time. The state-machine connection provides a foundation for these two choices: in the quantum model, unitary operations are notes and temporal boundaries are cues. An output map assigns sounds to designated readout-boundary/outcome pairs [28, 29]. Sounds are classical, but musical rules need not be.

A prescribed operation sequence defines a score, while a readout schedule determines when outcomes are recorded and sounded. With the current sound held between readouts, the schedule specifies a rhythm. Born-kernel composition determines which rhythms preserve the score's endpoint distribution.

Qubit rhythm laws. For an L-step score, choose a beat duration $\Delta > 0$. A rhythm is specified by an ordered tuple $\tau = (\ell_1, \dots, \ell_m)$ of positive integers summing to L, called an ordered composition of L. Its entries specify successive hold durations $\ell_1\Delta, \dots, \ell_m\Delta$, in chronological order from the entrance cue to the endpoint. The initial sound, specified separately from the quantum input state, is held from the entrance cue until the first readout. It may continue from the preceding passage even when the quantum state is freshly prepared. Put $t_0 = 0$ and $t_j = \sum_{r=1}^{j} \ell_r$; the schedule reads out at boundaries $t_1, \dots, t_m = L$. Its induced endpoint kernel is

$$\mathsf{K}_\tau(U_1, \dots, U_L) \coloneqq \mathsf{B}_\mathcal{C}(U_{t_m:t_{m-1}+1}) \circ \cdots \circ \mathsf{B}_\mathcal{C}(U_{t_1:1}). \tag{25}$$

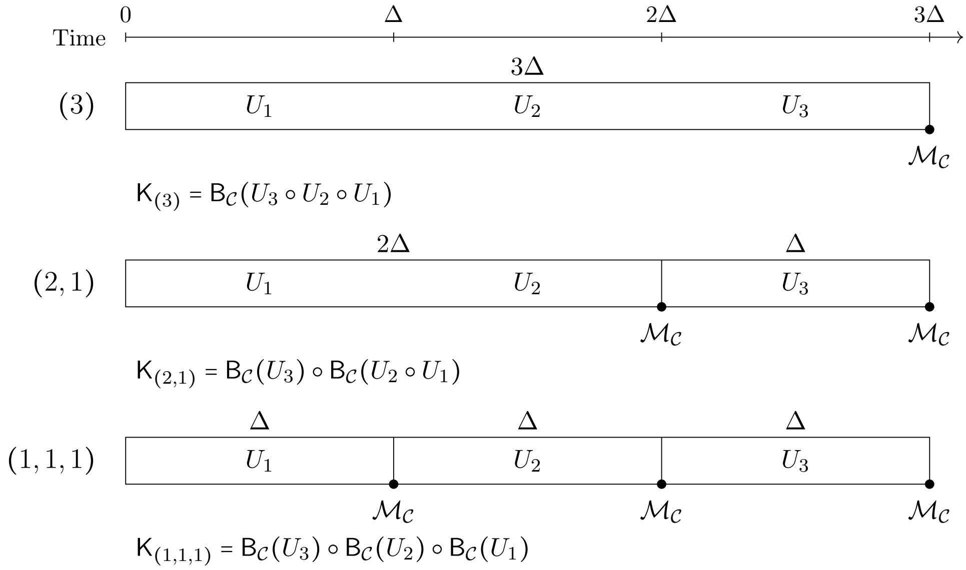
A rhythm τ is admissible when $\mathsf{K}_\tau = \mathsf{B}_\mathcal{C}(U_{L:1})$. We denote the set of admissible rhythms by $\mathscr{R}(U_1, \dots, U_L)$.

For the single-qubit passage in Eqs. (19)–(24), the admissible rhythms are:

Admissible rhythms of the qubit passage

$U_1 = R_{\sigma_y}(\pi/3), \qquad U_k = R_{\sigma_z}(\alpha_k) \circ R_{\sigma_x}(\pi/3), \quad k = 2, 3,$

$\alpha_2 = \arctan(1/2), \qquad \alpha_3 = \arctan\big(1/(2\sqrt{5})\big).$



A solid dot marks a readout $\mathcal{M}_{\mathcal{C}}$; block widths give the hold durations.
All three kernels equal $\mathsf{B}_{\mathcal{C}}(U_3 \circ U_2 \circ U_1)$, meaning that the sound distributions match at all three endpoints.

Evaluating Eq. (25) gives the following kernel comparison:

Endpoint-faithful schedules

$$\begin{aligned} \underbrace{\mathsf{B}_{\mathcal{C}}(U_3 \circ U_2 \circ U_1)}_{(3)} &= \underbrace{\mathsf{B}_{\mathcal{C}}(U_3) \circ \mathsf{B}_{\mathcal{C}}(U_2 \circ U_1)}_{(2,1)} \\ &= \underbrace{\mathsf{B}_{\mathcal{C}}(U_3) \circ \mathsf{B}_{\mathcal{C}}(U_2) \circ \mathsf{B}_{\mathcal{C}}(U_1)}_{(1,1,1)}, \end{aligned} \tag{26}$$

Excluded schedule

$$\mathsf{B}_{\mathcal{C}}(U_3 \circ U_2 \circ U_1) \neq \underbrace{\mathsf{B}_{\mathcal{C}}(U_3 \circ U_2) \circ \mathsf{B}_{\mathcal{C}}(U_1)}_{(1,2)}.$$

Every tuple in Eq. (26) is already in chronological musical order; the Born factors appear in reverse textual order because operator composition acts from right to left. Therefore, for the first three steps of the single-qubit passage in Eqs. (19) through (24), the set of admissible rhythms for this sequence is

$$\mathscr{R}(U_1, U_2, U_3) = \{(3), (2, 1), (1, 1, 1)\}. \tag{27}$$

This is not a restriction on all qubit scores. Membership in this set is controlled by relative phases across composite blocks. The admissible rhythm $(2, 1)$ reads out after U_2 and then U_3, whereas the excluded schedule $(1, 2)$ would read out after U_1 and then U_3. The latter leaves $U_3 \circ U_2$ as a coherent block beginning at an internal cue, where Born-kernel composition fails [Eq. (24)]. Removing the readout after U_2 from $(1, 1, 1)$ therefore gives the inadmissible schedule $(1, 2)$; removing the remaining intermediate readout restores admissibility by giving (3). For this score, reading out after every operation preserves the endpoint distribution of every prefix from the entrance cue. Restarting before U_2 means freshly preparing a context-basis input there and applying U_2 followed by U_3. A readout after U_2 then changes the final readout distribution after U_3. This is the quantum music paradox.

To sound the single-qubit passage, assign readout outcome 0 to the pitch E_2 and outcome 1 to the neighboring pitch F_2. This two-pitch realization, called the *Qubit Ostinato*, is described in Appendix F.1. The three faithful schedules articulate the same total span as one three-beat hold, a two-beat hold followed by a one-beat hold, or three one-beat holds; each enabled intermediate readout starts the assigned sound anew, even when its pitch is unchanged. Schedules outside $\mathscr{R}$ are physically possible but fail the Born-kernel composition condition.

Born-kernel composition therefore determines the admissible rhythms of the quantum passage. Intermediate outcomes may be retained and sounded: consistency concerns the terminal marginal, and marginalizing a record does not undo the measurement that created it. This supplies a concrete realization of Xenakis's "*minimum of logical constraints necessary for the construction of a musical process*" [20, p. 16]. As demonstrated below, one measurement of a selector qutrit chooses among exactly these three faithful schedules, so the rhythmic subdivision becomes a quantum-circuit output. Allowing arbitrary polynomial-size selector circuits with two musical output labels gives the prediction problem in Definition 5.1, which is PromiseBQP-complete by Theorem 5.2.

A compositional entangling score. Boundary stability means that every readout schedule is admissible on every contiguous passage of the score, for every context-basis input freshly prepared at its entrance (Proposition C.14). Appendix E gives an analytic three-step two-qubit family U_1, U_2, U_3 for which, at every temporal cut,

$$\begin{gathered} \mathsf{B}_{\mathcal{C}}(U_{b:a}) = \mathsf{B}_{\mathcal{C}}(U_b) \circ \cdots \circ \mathsf{B}_{\mathcal{C}}(U_a), \\ (1 \le a \le b \le 3). \end{gathered} \tag{28}$$

At $\xi = \eta = 1/6$ and $\zeta = 0$, every entry of every step is nonzero, all three steps are nonproduct perfect entanglers, and several nonzero paths reach every endpoint. Their interference terms do not vanish separately: Eq. (28) holds by aggregate cancellation.

A realization of the qubit-defined rhythms. Fix $\lambda = 2^{-1/3}$, set $\theta = \arccos\lambda$, and define

$$\begin{aligned} u_1 &= R_{\sigma_y}(\theta), \\ u_k &= R_{\sigma_z}(\alpha_k) \circ R_{\sigma_x}(\theta), \quad && k = 2, 3, \\ \tan\alpha_k &= \frac{\lambda^{k-1}\sqrt{1-\lambda^2}}{\sqrt{1-\lambda^{2(k-1)}}}, && k = 2, 3. \end{aligned}$$

Here $0 < \alpha_k < \pi/2$ fixes the phase branch. These are the first three single-qubit operations in Corollary C.11. Put $A_k = u_k \otimes u_k$, let $C_Z \coloneqq \mathrm{diag}(1, 1, 1, -1)$, and define the displayed two-qubit data steps by

$$U_1 = C_Z \circ A_1, \qquad U_2 = A_2 \circ C_Z, \qquad U_3 = C_Z \circ A_3.$$

The C_Z factors either cancel between adjacent steps or remain as computational-basis phases at a segment boundary, so they do not change the relevant Born kernels. Each U_k has full amplitude support and is locally equivalent to C_Z, hence is a perfect entangler. Prepare $|00\rangle$ on each run and use the computational-basis instrument whose non-selective action is $\mathcal{M}_{\mathcal{C}}(\rho) = \sum_{x=0}^{3} |x\rangle\langle x|\rho|x\rangle\langle x|$. The quantum state after outcome x is $|x\rangle$. The classical output map assigns a pitch using both the outcome and the readout boundary; rows Y_1, Y_2, Y_{end} refer to readouts after U_1, U_2, U_3, respectively:

	00	01	10	11
Y_1	E_4	G_4	C_4	C_5
Y_2	G_4	C_4	E_4	C_5
Y_{end}	C_4	E_4	G_4	C_5

For the fully checked rhythm $(1,1,1)$, the most probable pitches at the successive readouts are $\mathrm{E}_4, \mathrm{G}_4, \mathrm{C}_4$. All possible output pitches belong to the C-major triad, consisting of C, E, and G.

$$\begin{aligned}
\mathsf{B}_{\mathcal{C}}(U_3 \circ U_2 \circ U_1) &= \mathsf{B}_{\mathcal{C}}(U_3) \circ \mathsf{B}_{\mathcal{C}}(U_2 \circ U_1) \\
&= \mathsf{B}_{\mathcal{C}}(U_3) \circ \mathsf{B}_{\mathcal{C}}(U_2) \circ \mathsf{B}_{\mathcal{C}}(U_1) = [K(1/2)]^{\otimes 2}, \\
\Pr(Y_{\text{end}} \mid X = 00) &= \frac{1}{16}(9,3,3,1).
\end{aligned} \tag{29}$$

Thus the three selected readout rhythms have the same visibly biased endpoint law. Figure 4(c) also plots the excluded $(1,2)$ endpoint law from $|00\rangle$, approximately $(0.4032, 0.2318, 0.2318, 0.1332)$. For a recorded uniform entrance and a fair bit selecting the coherent or $(1,2)$ protocol, the corresponding schedule-information/current-size pair is $(\mathcal{I}(S; Y_{\text{end}} \mid X), \nu_{\mathcal{C}}(U_3 \circ U_2, U_1)) \simeq (0.0223, 0.2139)$, with information in bits per run.

Born-kernel composition determines the three admissible rhythms of this passage. A separate qutrit controller selects among them, with selection probabilities given by its own Born kernel. Before each data pass, apply V^{sel} to the qutrit prepared in its preceding observed state and measure it once, beginning with input $S = 0$. More generally, $V^{\text{sel}}(m)$ may be compiled from the prior audible history m, allowing previously observed sounds to control the selector between passages. In the selector basis $|0\rangle, |1\rangle, |2\rangle$, let J_3 be the all-ones matrix and define the displayed selector by

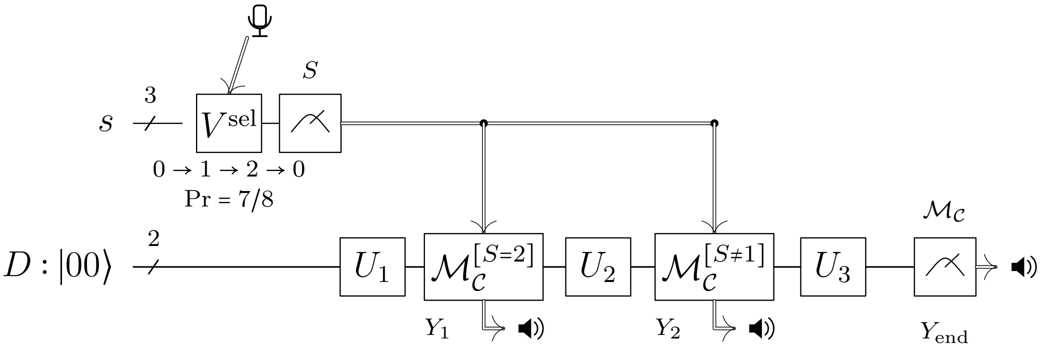
$$P_3|s\rangle = |(s+1) \bmod 3\rangle, \qquad a = \frac{-1 + \imath\sqrt{55}}{8},$$
$$V^{\text{sel}} = aP_3 + \frac{1}{4}(J_3 - P_3).$$

Its Born kernel is

$$\mathsf{B}_{\mathcal{C}_s}(V^{\text{sel}}) = \frac{1}{16}\begin{pmatrix} 1 & 1 & 14 \\ 14 & 1 & 1 \\ 1 & 14 & 1 \end{pmatrix}, \qquad \Pr\big(S_{r+1} = (S_r + 1) \bmod 3\big) = \frac{7}{8}. \tag{30}$$

All three selected rhythms have the same endpoint kernel. Changing their selection probabilities therefore changes the distribution of rhythms while preserving the endpoint distribution for every context-basis input. This remains true when the selector depends on earlier sound history, provided it selects a rhythm before the freshly prepared data pass.

For a proposition P, let $\mathcal{M}_{\mathcal{C}}^{[P]}$ denote $\mathcal{M}_{\mathcal{C}}$ when P is true and $\mathbb{1}$ otherwise. The measured value $S \in \{0, 1, 2\}$ controls the data pass



(31)

The upper wire carries the selector qutrit; after V^{sel}, its measurement supplies the classical value S. The double lines carry this value to the two conditional readouts on the lower data wire. The first is enabled only for $S = 2$, the second for $S = 0$ or 2, and the endpoint is always read out. Thus $S = 0, 1, 2$ select $(2, 1)$, (3), and $(1, 1, 1)$, respectively; the failing $(1, 2)$ mask is never selected. The microphone marks the optional prior sound history controlling V^{sel} between passes, and the speakers mark sounds assigned to readout outcomes. Every enabled readout starts its assigned sound anew, including when the pitch repeats. Every enabled intermediate readout retains its classical outcome, and the endpoint outcome Y_{end} is always sounded.

With adjacent circuit boundaries separated by Δ, the current sound is held until the next enabled readout, producing one-, two-, or three-Δ holds. The four pitches and three durations therefore define twelve pitch–duration symbols.

The displayed seed-1766 window in Figure 4 begins at an observed endpoint. The entrance sound carries over from that endpoint while the data state is freshly prepared in $|00\rangle$ for each pass. Five successive selector outcomes are $1, 2, 0, 1, 2$, and the five complete circuit passes occupy 15 beats. The mandatory final readout Y_{end} at $t = 15\Delta$ closes the fifth pass. Its assigned pitch C_4 is held until $t = 16\Delta$, followed by one silent beat ending at $t = 17\Delta$. This terminal hold is a playback convention after the measured passage and adds no quantum operation or readout. The figure also shows 256 separate eight-readout comparison excerpts; their sampled outcomes and durations are independent of this displayed terminal hold.

In the Walsh family, every readout schedule is admissible on every contiguous passage. For the displayed two-qubit score, precisely (3), $(2, 1)$, and $(1, 1, 1)$ are admissible on the full three-step passage, whereas $(1, 2)$ fails the Born-kernel composition condition. A score is cue-stable when the Born kernel of every independently initialized contiguous passage equals the ordered product of its step kernels, for every context-basis entrance. By Theorem 4.3, a cue-stable d-outcome score contains at most d full-support note operations,

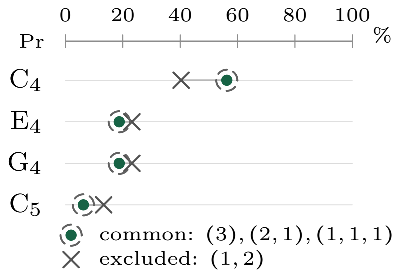
(a) Rhythm selection

S	Rhythm	Readouts after
0	$(2,1)$	U_2, then U_3
1	(3)	U_3
2	$(1,1,1)$	U_1, then U_2, then U_3

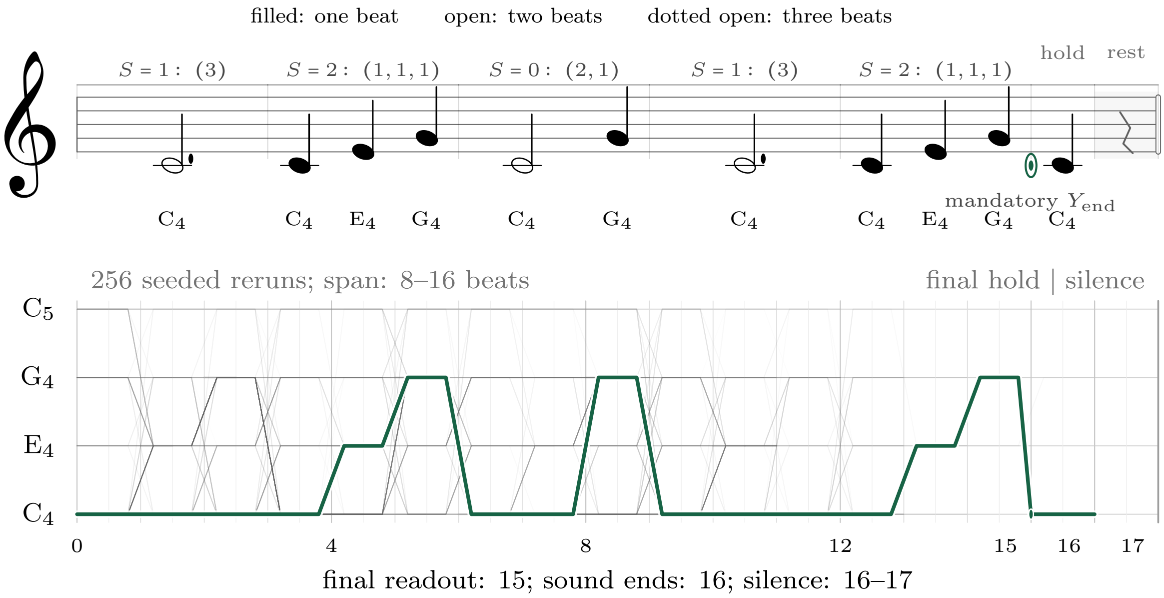
$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 0$

Each forward selector transition has probability 7/8.

(c)
Endpoint-law comparison



(b) Five complete passes, final sound, and one silent beat



(d) Circuit-sampled score realizations

Figure 4: **Quantum rhythm selection and its sounded realizations.** **(a)** The qutrit outcome in the main-text circuit, Eq. (31), selects one of the three admissible rhythms of the same U_1, U_2, U_3 passage. Every rhythm includes the final readout; the excluded schedule $(1,2)$ is never selected. **(b)** The seed-1766 performance begins at an observed endpoint. Five complete passes occupy beats 0–15, with selector outcomes $1,2,0,1,2$. Each pass begins in $|00\rangle$ while the preceding sound is held until the next readout. Filled, open, and dotted-open noteheads denote one-, two-, and three-beat holds, centered over their intervals. The circle at beat 15 marks the mandatory final readout; its C_4 is held until beat 16, followed by one silent beat to beat 17. This terminal hold adds no quantum operation or readout. **(c)** Green points inside dashed circles show the common endpoint law $(9,3,3,1)/16$ from $|00\rangle$ for (3), $(2,1)$, and $(1,1,1)$. Gray crosses show the excluded $(1,2)$ law, approximately $(0.4032, 0.2318, 0.2318, 0.1332)$; horizontal segments display the probability displacement. **(d)** Pale gray paths show 256 seeded simulator reruns, each retaining eight readouts and ending at its actual terminal beat. Their spans range from 8 to 16 beats. The highlighted performance has its final readout at 15, holds its final sound through 16, and is silent until 17. Ramps soften only the visual joins at readout boundaries; hold durations remain unchanged. Shared same-pitch background edges darken with their rerun count. The quantum measurement outcomes determine subsequent state preparation; the boundary-dependent pitch map assigns the classical sounds.

with equality attainable in every prime dimension.

Transition prediction. Given a circuit that generates the next musical output, the prediction question asks whether the probability of a designated output lies above an upper threshold or below a lower threshold, promised that one case holds. It concerns the output distribution across repeated preparations.

The computational model now ranges over arbitrary quantum circuits whose numbers of gates and qubits grow polynomially with the input length. The fixed qubit examples and the 3×3 qutrit selector illustrate the musical mechanisms. The completeness theorem concerns this growing circuit family, without imposing universal Born-kernel composition. Universally composing reversible state machines and tests of boundary stability retain the separate scopes established earlier.

Definition 5.1 (Quantum musical-transition prediction (QMTP)). Fix an integer $c \geq 1$. An instance consists of an explicitly encoded polynomial-size quantum circuit Q, initialized in $|0^n\rangle$, with a distinguished output qubit whose outcomes carry two fixed musical labels, and rational thresholds $0 \leq a < b \leq 1$. Let p_Q be the probability of the designated label and let $\ell := |\langle Q, a, b\rangle|$ be the total input bit length. The thresholds have inverse-polynomial separation $b - a \geq \ell^{-c}$. Decide whether

$$p_Q \geq b \qquad \text{or} \qquad p_Q \leq a, \tag{32}$$

promised that one alternative holds.

Replacing the two musical labels with ordinary machine output symbols leaves the probabilities, promise gap, and prediction problem unchanged.

Theorem 5.2 (PromiseBQP-completeness). *For every fixed $c \geq 1$, quantum musical-transition prediction (*QMTP*) is* PromiseBQP*-complete under deterministic classical polynomial-time many-one reductions.*

This completeness is inherited from standard circuit-output prediction: sampling relative to the thresholds' midpoint gives membership, while hardness maps a verifier's acceptance bit to the designated musical label [30, 31]. The complete gate-set conventions, encoding, and reduction are given in Appendix F.3.

The double-Hadamard passage is a workhorse of consistent-histories analysis [23, 24, 25, 26]. In the present fixed-context formulation it maximally violates Born-kernel composition, attaining the maximal normalized Born–Chapman–Kolmogorov current $\nu_{\mathcal{C}}(H, H) = 1$, and thereby connects this prediction problem directly to readout timing. Prepare $|0\rangle$, apply H followed by H, and always read out the endpoint. A separate circuit enables the intermediate readout when its output is $S = 1$, with probability p. With Y the final outcome,

$$\Pr(Y = 1 \mid S = 0) = 0, \qquad \Pr(Y = 1 \mid S = 1) = \frac{1}{2}, \qquad \Pr(Y = 1) = \frac{p}{2}.$$

Without the intermediate readout, $H^2 = \mathbb{1}$ returns the prepared state; with it, the final outcome is uniform. The readout choice selects the rhythms (2) and $(1, 1)$, whose endpoint kernels differ for this passage. Assigning distinct sounds to $Y = 0, 1$ makes the endpoint probability an audible-output probability. The input thresholds a, b become $a/2, b/2$, so their separation remains inverse polynomial. The resulting promise problem is PromiseBQP-complete by Corollary F.6. Here the endpoint distribution depends on rhythm selection; the three admissible rhythms in Figure 4 share one endpoint distribution.

If quantum musical-transition prediction were efficiently classically predictable, then BQP = BPP.

Experimental outlook. The normalized magnitude $\nu_{\mathcal{C}}(V, U)$ of the composition defect is operationally estimable. For each context-basis input, one compares endpoint frequencies from two executions of the same two-step passage, differing only by the insertion of an unread context measurement at the intermediate boundary; the resulting kernels determine $\nu_{\mathcal{C}}$ through their normalized Frobenius separation. For the saturating Hadamard pair of Theorem D.4, a sequence of two balanced beam splitters gives $K_{\mathrm{end}} = \mathbb{1}$ and $K_{\mathrm{step}} = J_2/2$, hence $\nu_{\mathcal{C}}(H, H) = 1$: the endpoint-only protocol returns either prepared alternative with certainty, whereas the stepwise protocol is uniform [5]. More generally, for a fixed context, a sequence is boundary-stable precisely when these endpoint kernels agree for every context-basis input and every internal cut of every contiguous subpassage. Mid-circuit measurement has been demonstrated in superconducting and trapped-ion processors [38, 39], making direct estimation of $\nu_{\mathcal{C}}$ experimentally feasible. Theorem 5.2 addresses the distinct prediction problem: an efficient classical algorithm for predicting the designated musical output of arbitrary quantum circuits would imply BQP = BPP.

6 Conclusion

The quantum–classical interface is rich: unread intermediate measurements can change the quantum state without changing the endpoint statistics, while genuinely mixing sequences can reproduce their stochastic transition law at every prefix for arbitrarily many steps. That agreement can nevertheless fail when the state is freshly prepared at an internal boundary and the unchanged remaining operations are applied. Endpoint fidelity from the preparation boundary need not supply a reusable stochastic law on proper subintervals.

Boundary stability means that the operation-supplied transition laws must remain valid on every independently initialized contiguous passage. This demand has a sharp finite-dimensional cost: at most d full-support steps in dimension d, with the bound attained in every prime dimension. The dense two-qubit realization shows that exact composition can survive entangled dynamics with active interference.

The Born–Chapman–Kolmogorov current records the composition defect, and its normalized Frobenius norm makes the failure quantitative. The associated conditional schedule information measures, in bits per run and conditioned on the chosen input, what the endpoint reveals about a fair choice between coherent and checked protocols; it vanishes exactly when their endpoint laws agree.

The state-machine connection provides a foundation for a quantum theory of music: unitary operations are notes, temporal boundaries are cues, and a classical output map assigns sounds to readout outcomes. Born-kernel composition determines which readout rhythms preserve a score's endpoint distribution. A score can preserve that distribution under stepwise readout from its entrance cue yet fail the corresponding test after fresh preparation at an internal cue. This is the quantum music paradox. For the circuit-universal score model, promised designated-transition prediction is PromiseBQP-complete.

Our question remains narrower than the general measurement problem, but identifies a common boundary between measurement-induced consistency, stochastic processes, and computation. This framework introduces broad classes of quantum state machines.

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Popular summary.

The problem.
Quantum probabilities for a multistep process generally cannot be reconstructed by multiplying step-by-step transition probabilities: checking the intermediate steps can change the final outcome statistics.

The question.
When can a sequence of unitary operations be measured at every intermediate step without changing the final outcome distribution? How badly can this agreement fail otherwise?

The answer.
We classify when stepwise and endpoint statistics agree when measured on every contiguous interval. We then introduce a normalized measure of their disagreement, prove its bound is sharp, and prove that deciding whether a designated musical output probability lies above or below promised thresholds is PromiseBQP-complete. The prediction result also applies to ordinary machine output symbols.

The consequence.
Applied to quantum music[a], unitary operations are notes and temporal boundaries are cues. Recorded outcomes produce sounds, and the times between readouts determine how long those sounds are held. Composition identifies the rhythms that preserve the final outcome distribution. Reading out after every operation can preserve that distribution from the opening cue, yet change it when the unchanged remaining operations begin from a fresh preparation at a later cue. This is the quantum music paradox.

[a]Interactive project page with musical demonstrations: `https://institut-kets.github.io/quantum-composition/#composition-heading`.

Appendix to "The Quantum Composition Paradox"

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This appendix is organized around verification of the main paper's claims. Compact conventions and a proof map are followed by the universal classification and its state-machine consequence, prescribed-pair and boundary results, quantitative bounds, and the analytic circuit. Musical realizations and transition-prediction complexity come last, so their mathematical foundations are available before the application.

Contents

A Conventions and proof map

Section B proves the universal classification and its reversible-state-machine consequence. Section C develops the prescribed-pair criteria, prefix and boundary laws, and their relation to dephasing and histories. Section D gives the schedule-information, current, and capacity bounds. Section E verifies the analytic three-step circuit. Section F supplies the musical realizations and the prediction and sampling proofs.

A.1 Fixed context and chronological conventions

Remark A.1 (Standard notation). We write $\imath := \sqrt{-1}$, $\mathbb{1}$ for the identity, Tr for the trace, and $\mathrm{U}(d)$ for the unitary group. The symbols $A^\dagger$, $\overline{A}$, and A^T denote conjugate transpose, entrywise conjugation, and transpose. We use the norms

$$\|A\|_\mathrm{F} := [\mathrm{Tr}(A^\dagger A)]^{1/2}, \qquad \|A\|_\mathrm{op} := \sup_{\|v\|_2=1} \|Av\|_2.$$

Definition A.2 (Finite fixed-context model). Fix $\mathsf{H} \cong \mathbb{C}^d$, $d \geq 2$, the outcome set $X = \{0, \dots, d-1\}$, and the atomic context $\mathcal{C} = \{P_x = |x\rangle\langle x| : x \in X\}$ of an orthonormal basis, with $\sum_x P_x = \mathbb{1}$. A $\mathcal{C}$-diagonal preparation is $\rho = \sum_x q(x) P_x$, where q is a probability distribution. The Born kernel of $U \in \mathrm{U}(d)$ and the unread context measurement are

$$[\mathsf{B}_\mathcal{C}(U)]_{yx} := \mathrm{Tr}(P_y U P_x U^\dagger) = |\langle y|U|x\rangle|^2, \qquad \mathcal{M}_\mathcal{C}(\rho) := \sum_x P_x \rho P_x.$$

We abbreviate $\mathsf{B}_\mathcal{C}$ to B when the context is unambiguous. The Born kernel acts on outcome probability vectors.

Definition A.3 (Stochastic kernels and chronological composition). A stochastic kernel on X is a real matrix $K = (K_{yx})$ with $K_{yx} \geq 0$ for all x, y and $\sum_y K_{yx} = 1$ for every input x. Columns label inputs, rows label outputs, and column probability vectors evolve as $q' = Kq$. A kernel is *doubly stochastic* (also called *bistochastic*) when its rows also sum to one and *unistochastic* when $K = \mathsf{B}_\mathcal{C}(U)$ for some unitary U. Throughout, $V \circ U$ means that U acts before V; the same chronological convention is used for unitary operations and stochastic kernels.

Definition A.4 (Passages, boundaries, and restarts). A prescribed passage is an ordered finite sequence $(U_1, \dots, U_L) \in \mathrm{U}(d)^{\times L}$, with U_1 acting first. For $1 \leq a \leq b \leq L$, write

$$U_{b:a} := U_b \circ U_{b-1} \circ \cdots \circ U_a.$$

for the contiguous passage containing the steps $U_a, \dots, U_b$, with both endpoint steps included; in particular, $U_{k:k} = U_k$. Set $K_{b:a} := \mathsf{B}_\mathcal{C}(U_{b:a})$. Number the temporal boundaries $0, \dots, L$, so that U_k carries boundary $k-1$ to boundary k and $U_{b:a}$ begins at boundary $a-1$ and ends at boundary b. A *restart immediately before* U_a, at boundary $a-1$, means preparing a fresh $\mathcal{C}$-diagonal state there and applying $U_{b:a}$; it does not mean carrying forward the quantum state from an earlier passage. In the musical semantics, every temporal boundary is a *cue*; a restart specifies a fresh preparation at the chosen cue.

Remark A.5 (Step-index check). A three-step passage has boundaries $0, 1, 2, 3$:

$$U_{3:1} = U_3 \circ U_2 \circ U_1, \qquad U_{3:2} = U_3 \circ U_2.$$

This inclusive step-index convention is used throughout the paper and appendix.

A.2 Composition scopes

Definition A.6 (Composition scopes and operation-supplied genealogy). An interval family $\mathscr{I}$ consists of inclusive step intervals $[a, b]$. A cut after c, with $a \le c < b$, is allowed when $[a, b], [a, c], [c+1, b] \in \mathscr{I}$. The actual interval Born kernels carry a *Born-induced stochastic genealogy*, or *operation-supplied genealogy*, when

$$K_{b:a} = K_{b:c+1} \circ K_{c:a} \qquad \text{at every allowed cut.} \tag{33}$$

A sequence is *prefix-consistent* when

$$\mathsf{B}_{\mathcal{C}}(U_{k:1}) = \mathsf{B}_{\mathcal{C}}(U_k) \circ \cdots \circ \mathsf{B}_{\mathcal{C}}(U_1) \qquad (2 \le k \le L), \tag{34}$$

and *boundary-stable* when

$$\begin{gathered} \mathsf{B}_{\mathcal{C}}(U_{b:a}) = \mathsf{B}_{\mathcal{C}}(U_b) \circ \cdots \circ \mathsf{B}_{\mathcal{C}}(U_a), \\ (1 \le a \le b \le L). \end{gathered} \tag{35}$$

Boundary stability is equivalent to a genealogy on all contiguous step intervals; prefix consistency is weaker. In the musical semantics, boundary stability is *cue stability* under fresh preparation at any cue.

Remark A.7 (Four composition scopes). A pair identity concerns two adjacent blocks; prefix consistency concerns every passage from boundary 0; boundary stability concerns every contiguous passage. Universal composition quantifies over every ordered pair in a composition-closed family (Section B). An identity only for the complete passage's endpoint kernel does not impose prefix consistency.

Definition A.8 (Support and qubit mixing). A stochastic kernel has *full support* when every entry is strictly positive. For qubit Born kernels only, *genuinely mixing* means full support. We reserve *strictly positive* for an entrance distribution π satisfying $\pi(x) > 0$ for every x. A *unitary monomial* is a product $\Delta \circ \Pi$ of a diagonal unitary Δ and a permutation matrix Π; phase-dressed permutation is its descriptive synonym.

Remark A.9 (Register and qubit notation). We reserve L for the number of passage steps and n for the number of qubits. An n-qubit register has $d = N := 2^n$ computational-basis outcomes and wires $q_0, \ldots, q_{n-1}$. For the Pauli matrices $\sigma_x, \sigma_y, \sigma_z$, we use $R_\sigma(\theta) := e^{-\imath\theta\sigma/2}$.

Remark (context-diagonal preparations). For an entrance probability vector p and the preparation $\rho = \sum_x p_x P_x$, the context-outcome distribution after U is

$$q_y = \mathrm{Tr}(P_y U \rho U^\dagger) = \sum_x [\mathsf{B}_{\mathcal{C}}(U)]_{yx} p_x, \qquad q = \mathsf{B}_{\mathcal{C}}(U) p.$$

Thus two endpoint kernels agree if and only if they give the same output distribution for every context-basis preparation, equivalently for every $\mathcal{C}$-diagonal preparation. The displayed equality follows from linearity in ρ. Kernel equality implies equality on every probability vector. Conversely, the preparations $p_x = \delta_{xz}$ test each column z separately. The same argument applies to products of Born kernels induced by readout schedules.

A.3 Born kernels and the composition current

Classical path probabilities multiply, and marginalizing an intermediate label composes the adjacent kernels by the Chapman–Kolmogorov rule [1, 2]. The current measures the failure of this rule for the Born kernels of prescribed unitary passages.

Definition A.10 (Born–Chapman–Kolmogorov current). For U followed by V, the endpoint-checked and stepwise-checked kernels are

$$\begin{aligned} K_{\text{end}} &= \mathsf{B}(V \circ U), \\ K_{\text{step}} &= \mathsf{B}(V) \circ \mathsf{B}(U). \end{aligned} \tag{36}$$

Here *endpoint-checked* means that the passage remains coherent until one final context readout; *stepwise-checked* inserts the unread measurement channel $\mathcal{M}_{\mathcal{C}}$ at each prescribed intermediate boundary. Their signed difference

$$\mathcal{J}_{\mathcal{C}}(V, U) := K_{\text{end}} - K_{\text{step}} \tag{37}$$

becomes a dimensionless signed Born-composition defect, which we also call the Born–Chapman–Kolmogorov current. Its columns sum to zero. It vanishes exactly when an unread context check after U preserves every final context probability for every basis preparation.

Definition A.11 (Normalized Frobenius norm of the current). Assume $d \geq 2$. For a $d \times d$ complex matrix $M = (M_{yx})_{x,y \in X}$, the Frobenius norm is the Euclidean length of the vector formed by its d^2 entries:

$$\|M\|_{\mathrm{F}} := \left(\sum_{x,y \in X} |M_{yx}|^2 \right)^{1/2} = \left[\mathrm{Tr}(M^\dagger M) \right]^{1/2}. \tag{38}$$

Applying Eq. (38) to the current, we define its normalized Frobenius norm by

$$\nu_{\mathcal{C}}(V, U) := \frac{1}{\sqrt{d-1}} \, \|\mathcal{J}_{\mathcal{C}}(V, U)\|_{\mathrm{F}}. \tag{39}$$

The factor $\sqrt{d-1}$ is the sharp maximum of the current's Frobenius norm, proved in Theorem D.4. Consequently $0 \leq \nu_{\mathcal{C}}(V, U) \leq 1$, and $\nu_{\mathcal{C}}(V, U) = 0$ exactly when the current vanishes.

B Universal classification and reversible state machines

B.1 Universal composition

Definition B.1 (Unitary grammar and universal composition). A *unitary grammar* Γ is a composition-closed subsemigroup of $\mathrm{U}(d)$ generated by the allowed operations. Universal composition means that the canonical Born map is a homomorphism on every ordered pair:

$$\mathsf{B}_{\mathcal{C}}(V \circ U) = \mathsf{B}_{\mathcal{C}}(V) \circ \mathsf{B}_{\mathcal{C}}(U) \tag{40}$$

for every $U, V \in \Gamma$. This quantifier is stronger than the genealogy law for one prescribed interval family in Eq. (33).

B.2 Classification theorem

Theorem B.2 (Classification of Born-consistent unitary semigroups). *Let $\Gamma \subseteq \mathrm{U}(d)$ be closed under composition. Then*

$$\mathsf{B}_{\mathcal{C}}(V \circ U) = \mathsf{B}_{\mathcal{C}}(V) \circ \mathsf{B}_{\mathcal{C}}(U) \qquad \textit{for all } U, V \in \Gamma$$

if and only if every element of Γ is a unitary monomial in the context basis.

B.3 Bistochastic contractions

Lemma B.3 (Bistochastic contraction). *Every real $d \times d$ bistochastic matrix M is a contraction in Euclidean norm:*

$$\|Mv\|_2 \le \|v\|_2 \qquad (v \in \mathbb{R}^d). \tag{41}$$

If M is an isometry, then M is a permutation matrix.

Proof. Jensen's inequality applied to each row gives

$$\|Mv\|_2^2 = \sum_y \left(\sum_x M_{yx} v_x \right)^2 \le \sum_y \sum_x M_{yx} v_x^2 \tag{42}$$

$$= \sum_x v_x^2 \sum_y M_{yx} = \|v\|_2^2. \tag{43}$$

If equality holds for every v, then $M^{\top} M = \mathbb{1}$. Each column of M is nonnegative, has sum one, and has squared Euclidean norm one. A probability vector has squared norm one only when one entry is one and all others vanish. The row sums place these unit entries in distinct rows, so M is a permutation matrix. □

B.4 Lemmas for the semigroup theorem

Lemma B.4 (Unitary recurrence). *For every finite-dimensional unitary U, there is a sequence $m_j \to \infty$ for which $U^{m_j} \to \mathbb{1}$.*

Proof. Write the eigenvalues of U as $e^{2\pi\imath\theta_1}, \ldots, e^{2\pi\imath\theta_d}$. Simultaneous Diophantine approximation supplies integers $m_j \to \infty$ for which every distance $\|m_j\theta_r\|_{\mathbb{R}/\mathbb{Z}}$ tends to zero. Hence each eigenvalue of U^{m_j} tends to one, and therefore $U^{m_j} \to \mathbb{1}$ in operator norm. □

Lemma B.5 (Recurrent bistochastic powers). *Let A be a real bistochastic matrix. If $A^{m_j} \to \mathbb{1}$ along a sequence $m_j \to \infty$, then A is a permutation matrix.*

Proof. For every $v \in \mathbb{R}^d$, Lemma B.3 gives

$$\|v\|_2 = \lim_j \|A^{m_j} v\|_2 \le \|Av\|_2 \le \|v\|_2.$$

Thus A is an isometry, and the final assertion of Lemma B.3 makes it a permutation matrix. □

B.5 Proof of the classification of Born-consistent unitary semigroups

Proof of Theorem B.2. Assume the universal composition identity in Theorem B.2 and choose $U \in \Gamma$. Put $A = \mathsf{B}_{\mathcal{C}}(U)$. Closure and induction give

$$\mathsf{B}_{\mathcal{C}}(U^m) = A^m \qquad (m \ge 1). \tag{44}$$

By Lemma B.4, choose $m_j \to \infty$ with $U^{m_j} \to \mathbb{1}$. Continuity of the Born map and Eq. (44) give $A^{m_j} \to \mathbb{1}$, so Lemma B.5 makes A a permutation matrix. Since $|U_{yx}|^2 = A_{yx}$, the unitary U has one nonzero entry of unit modulus in each row and column. Thus U is a unitary monomial, that is, a phase-dressed permutation.

Conversely, write $U = \Delta_U \circ \Pi_U$ and $V = \Delta_V \circ \Pi_V$, where the Δ matrices are diagonal unitary and the Π matrices are permutations. Then

$$\mathsf{B}_{\mathcal{C}}(U) = \Pi_U, \qquad \mathsf{B}_{\mathcal{C}}(V) = \Pi_V, \qquad \mathsf{B}_{\mathcal{C}}(V \circ U) = \Pi_V \circ \Pi_U, \tag{45}$$

which proves the converse direction of Theorem B.2. □

Equivalently, these are precisely the unitaries that preserve incoherence relative to $\mathcal{C}$: the normalizer of the diagonal algebra [25].

B.6 Universal composition forces reversible state machines

Definition B.6 (Moore machine and fixed-context unitary completion). A Moore machine is a tuple $\mathfrak{M} = (X, \Sigma, \delta, \lambda_{\text{out}}, x_0)$, where X is a finite state set, Σ is a finite input alphabet, $\delta : X \times \Sigma \to X$ is the transition function, $\lambda_{\text{out}} : X \to \mathsf{A}$ is an output map to a finite alphabet, and $x_0 \in X$ is the initial state. A fixed-context unitary completion assigns $U_\sigma \in \mathrm{U}(d)$ to each $\sigma \in \Sigma$ so that $[\mathsf{B}_{\mathcal{C}}(U_\sigma)]_{yx} = 1$ exactly when $y = \delta(x, \sigma)$. The completion is universal when the generated composition-closed family obeys universal Born-kernel composition.

Corollary B.7 (Universal completion is exactly reversible). *A finite Moore machine admits a universal fixed-context unitary completion if and only if every map $\delta_\sigma : x \mapsto \delta(x, \sigma)$*

is bijective. Equivalently, universal completion uniquely determines reversible deterministic transitions, although their diagonal unitary phases remain arbitrary.

Proof. Universal Born-kernel composition uniquely determines a reversible classical transition for every input symbol. Let X label the outcomes of the atomic context $\mathcal{C}$, let Σ be a finite input alphabet, assign a unitary U_σ to each $\sigma \in \Sigma$, and let

$$\Gamma = \langle U_\sigma : \sigma \in \Sigma \rangle \tag{46}$$

be the composition-closed family generated by those operations. If

$$\mathsf{B}_{\mathcal{C}}(V \circ U) = \mathsf{B}_{\mathcal{C}}(V) \circ \mathsf{B}_{\mathcal{C}}(U) \qquad \text{for all } U, V \in \Gamma \tag{47}$$

holds, Theorem B.2 gives, for every $\sigma \in \Sigma$, a unique permutation π_σ of X with

$$\mathsf{B}_{\mathcal{C}}(U_\sigma) = \Pi_\sigma, \qquad (\Pi_\sigma)_{yx} = \begin{cases} 1, & y = \pi_\sigma(x), \\ 0, & y \neq \pi_\sigma(x), \end{cases} \tag{48}$$

Define $\delta(x, \sigma) := \pi_\sigma(x)$ and write $\delta_\sigma(x) := \delta(x, \sigma)$. For any finite output alphabet A, output map $\lambda_{\mathrm{out}} : X \to \mathsf{A}$, and initial state $x_0 \in X$, the induced Moore machine $\mathfrak{M} = (X, \Sigma, \delta, \lambda_{\mathrm{out}}, x_0)$ is therefore deterministic and reversible: every $\delta_\sigma : X \to X$ is a bijection.

Conversely, a preassigned Moore machine admits a universal fixed-context unitary completion, meaning $[\mathsf{B}_{\mathcal{C}}(U_\sigma)]_{yx} = 1$ exactly when $y = \delta(x, \sigma)$, if and only if every δ_σ is bijective. For a reversible machine, take the permutation matrix Π_σ of δ_σ and choose $U_\sigma = \Delta_\sigma \circ \Pi_\sigma$ with arbitrary diagonal unitaries Δ_σ. The output map λ_{out} and initial state x_0 play no role in this classification. The phases make the unitary completion nonunique but remain invisible to its uniquely induced fixed-context transition.

□

C Prescribed pairs and boundary results

We first express pair composition as a compression identity and then give the qubit and qutrit criteria. Prefix recurrence and the path formulation lead to the boundary and schedule conditions, followed by the distinction from consistent fine-grained histories.

C.1 The double-operator Born-kernel linearization

Definition C.1 (Double-operator representation and sector projections). Let $\mathsf{H}_d \cong \mathbb{C}^d$ be a separate Hilbert space over the classical labels, and let $\mathrm{HS}(\mathsf{H}) := \mathrm{End}(\mathsf{H})$ denote the complex vector space of linear operators on H, equipped with the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{Tr}(A^\dagger B)$. Define

$$\iota_{\mathcal{C}} : \mathsf{H}_d \longrightarrow \mathrm{HS}(\mathsf{H}), \qquad \iota_{\mathcal{C}} |x\rangle_{\mathrm{cl}} := P_x. \tag{49}$$

The projectors P_x are orthonormal, so $\iota_{\mathcal{C}}^{\dagger}\iota_{\mathcal{C}} = \mathbb{1}$. The adjoint action of a unitary is the double-operator action

$$L_U : \mathrm{HS}(\mathsf{H}) \longrightarrow \mathrm{HS}(\mathsf{H}), \qquad L_U(A) = UAU^{\dagger}. \tag{50}$$

With column-stacking vectorization in the context basis, L_U is represented by $U \otimes \overline{U}$. The Born kernel is its contraction to the diagonal sector:

$$\mathsf{B}_{\mathcal{C}}(U) = \iota_{\mathcal{C}}^{\dagger} \circ L_U \circ \iota_{\mathcal{C}}. \tag{51}$$

Let

$$P_{\mathcal{C}} := \iota_{\mathcal{C}} \circ \iota_{\mathcal{C}}^{\dagger}, \qquad Q_{\mathcal{C}} := \mathbb{1} - P_{\mathcal{C}}. \tag{52}$$

The two projections select the diagonal and off-diagonal Hilbert–Schmidt sectors.

Proposition C.2 (Double-operator composition identity). *For all $U, V \in \mathrm{U}(d)$,*

$$\mathsf{B}_{\mathcal{C}}(V \circ U) = \mathsf{B}_{\mathcal{C}}(V) \circ \mathsf{B}_{\mathcal{C}}(U) + \iota_{\mathcal{C}}^{\dagger} \circ L_V \circ Q_{\mathcal{C}} \circ L_U \circ \iota_{\mathcal{C}}. \tag{53}$$

Proof. Use $L_{V\circ U} = L_V \circ L_U$ and insert $\mathbb{1} = P_{\mathcal{C}} + Q_{\mathcal{C}}$ between the lifted operations. Since $P_{\mathcal{C}} = \iota_{\mathcal{C}} \circ \iota_{\mathcal{C}}^{\dagger}$, the diagonal term is the product of the two contracted Born kernels, and the off-diagonal term is the displayed correction. □

The defect is therefore the coherence-mediated correction that leaves the diagonal population sector under U and returns to that sector under V. Quantum composition itself is unchanged; the mismatch is created by contracting to that sector between operations. Equivalently, on Hilbert–Schmidt space,

$$P_{\mathcal{C}} \circ L_V \circ L_U \circ P_{\mathcal{C}} = P_{\mathcal{C}} \circ L_V \circ P_{\mathcal{C}} \circ L_U \circ P_{\mathcal{C}} + P_{\mathcal{C}} \circ L_V \circ Q_{\mathcal{C}} \circ L_U \circ P_{\mathcal{C}}. \tag{54}$$

The correction is coherence created by U and read back into populations by V. It is a fixed-context projection effect, not an open-system memory kernel; compare to the projection-operator literature [23, 24].

C.2 The scalar form of every qubit Born kernel

Definition C.3 (Qubit Born-kernel coordinate). For $-1 \le \lambda \le 1$, define

$$K(\lambda) := \frac{1+\lambda}{2}\mathbb{1} + \frac{1-\lambda}{2}\sigma_x.$$

Every 2×2 bistochastic matrix has this form, and $K(\mu) \circ K(\lambda) = K(\mu\lambda)$. For a qubit unitary U, its population polarization is

$$\beta(U) := \langle 0|U^{\dagger}\sigma_z U|0\rangle, \qquad \mathsf{B}_{\mathcal{C}}(U) = K(\beta(U)).$$

The kernel has full support, and is therefore genuinely mixing in the terminology of Definition A.8, exactly when $|\lambda| < 1$.

Write

$$\begin{aligned}\sigma_x &= |0\rangle\langle 1| + |1\rangle\langle 0|,\\ \sigma_y &= -\imath|0\rangle\langle 1| + \imath|1\rangle\langle 0|,\\ \sigma_z &= |0\rangle\langle 0| - |1\rangle\langle 1|.\end{aligned} \tag{55}$$

In the computational context, Dirac notation gives

$$\mathsf{B}(U) = \begin{pmatrix} |\langle 0|U|0\rangle|^2 & |\langle 0|U|1\rangle|^2 \\ |\langle 1|U|0\rangle|^2 & |\langle 1|U|1\rangle|^2 \end{pmatrix} = K(\beta(U)), \tag{56}$$

where

$$\beta(U) := \langle 0|U^\dagger \sigma_z U|0\rangle = |\langle 0|U|0\rangle|^2 - |\langle 1|U|0\rangle|^2. \tag{57}$$

Thus

$$K(\lambda) := \frac{1+\lambda}{2}\mathbb{1} + \frac{1-\lambda}{2}\sigma_x. \tag{58}$$

Every 2×2 bistochastic matrix has this form, and

$$K(\mu)\circ K(\lambda) = K(\mu\cdot\lambda). \tag{59}$$

Here λ is a generic kernel parameter, whereas $\beta(U)$ is the unitary-specific polarization. The former is the nontrivial eigenvalue of $K(\lambda)$ on the signed population mode $(1,-1)^\mathsf{T}$. The kernel has full support exactly when $|\lambda| < 1$;

Definition A.8 calls this genuine mixing and also fixes the equivalent unitary-monomial terminology. Therefore

$$\mathsf{B}(V\circ U) = \mathsf{B}(V)\circ\mathsf{B}(U) \quad\Longleftrightarrow\quad \beta(V\circ U) = \beta(V)\beta(U). \tag{60}$$

Definition C.4 (Qubit interference scalar). For two paths through the intermediate context, define the real interference scalar

$$\chi(V,U) := 4\,\mathrm{Re}\Big[\langle 0|V|0\rangle\langle 0|U|0\rangle\overline{\langle 0|V|1\rangle\langle 1|U|0\rangle}\Big]. \tag{61}$$

Theorem C.5 (Classification of pairs of one-qubit unitary steps). *For arbitrary qubit unitaries U, V,*

$$\mathsf{B}(V\circ U) = \mathsf{B}(V)\circ\mathsf{B}(U) \quad\Longleftrightarrow\quad \chi(V,U) = 0, \tag{62}$$

and

$$\mathsf{B}(V\circ U) - \mathsf{B}(V)\circ\mathsf{B}(U) = \frac{\chi(V,U)}{2}(\mathbb{1} - \sigma_x). \tag{63}$$

Because $\|\mathbb{1} - \sigma_x\|_\mathrm{F} = 2$, the normalized current introduced in Eq. (39) reduces for a qubit to $\nu_\mathcal{C}(V,U) = |\chi(V,U)|$. Thus the scalar classification is exactly the zero set of the current used in the main text.

Proof. For entrance and endpoint 0, the coherent amplitude is the sum of the two context-path amplitudes

$$\langle 0|V\circ U|0\rangle = \langle 0|V|0\rangle\langle 0|U|0\rangle + \langle 0|V|1\rangle\langle 1|U|0\rangle. \tag{64}$$

The difference between the squared modulus of this sum and the sum of the two squared moduli is $\chi(V,U)/2$. Both kernels are 2×2 bistochastic, so their signed difference has zero row and column sums and is consequently this single scalar times $\mathbb{1}-\sigma_x$. This proves both claims. $\square$

The same scalar identity has a direct path-amplitude form. Resolve the intermediate context identity and define

$$a_j(V,U) := \langle 0|V|j\rangle\langle j|U|0\rangle, \qquad j\in\{0,1\}. \tag{65}$$

Then

$$\langle 0|V\circ U|0\rangle = \sum_{j=0}^{1}\langle 0|V|j\rangle\langle j|U|0\rangle = a_0(V,U)+a_1(V,U), \tag{66}$$

and

$$\chi(V,U) = 4\,\mathrm{Re}\Big[a_0(V,U)\overline{a_1(V,U)}\Big]. \tag{67}$$

Equation (67) already shows that the composition law holds precisely when the two path amplitudes have vanishing real cross term.

The body witness follows immediately. For $U = R_{\sigma_y}(\pi/3)$ and $V = R_{\sigma_x}(\phi)$ with $\cos\phi = 1/3$, the two path amplitudes are

$$a_0(V,U) = \frac{\sqrt{2}}{2}, \qquad a_1(V,U) = -\frac{\imath}{2\sqrt{3}}.$$

Both are nonzero and have relative phase $-\pi/2$, so their interference term has zero real part. Consequently,

$$\mathsf{B}(U) = K(1/2), \qquad \mathsf{B}(V) = K(1/3), \qquad \mathsf{B}(V\circ U) = K(1/6). \tag{68}$$

More generally, for $U = R_{\sigma_y}(\theta)$ and $V = R_{\sigma_x}(\phi)$, the two path amplitudes are respectively real and purely imaginary. Therefore

$$\mathsf{B}(V\circ U) = K(\cos\phi\cos\theta) = \mathsf{B}(V)\circ\mathsf{B}(U). \tag{69}$$

This continuous family includes pairs with unequal, genuinely mixing local kernels, so equality of the two individual Born kernels is neither required nor generic.

The defect therefore vanishes in two ways. One path amplitude may be zero, in which case at least one of the qubit unitaries is a unitary monomial. Otherwise the two path amplitudes are both nonzero but differ in phase by $\pm\pi/2$, so their conjugate cross terms cancel exactly.

C.3 Qutrit two-step analysis

The scalar kernel $K(\lambda)$ is special to a qubit. A generic qutrit Born kernel lies in the four-dimensional set of 3×3 unistochastic matrices, and phases invisible to an individual Born kernel can reappear when two unitaries are composed. Thus there is no single-parameter qutrit law analogous to Eq. (59). There is, however, a necessary-and-sufficient geometric criterion for every prescribed ordered pair of qutrit unitary steps (U,V).

In this subsection, (U, V) denotes an ordered pair of qutrit unitary steps, with U acting before V.

Definition C.6 (Qutrit population–coherence blocks and frames). Let $\mathsf{Herm}_0(3)$ be the real Hilbert space of traceless Hermitian 3×3 matrices with the Hilbert–Schmidt inner product. The fixed context gives the orthogonal decomposition

$$\mathsf{Herm}_0(3) = \mathsf{D} \oplus \mathsf{C}_{\mathrm{coh}}, \qquad \dim_{\mathbb{R}} \mathsf{D} = 2, \qquad \dim_{\mathbb{R}} \mathsf{C}_{\mathrm{coh}} = 6, \tag{70}$$

where D is the diagonal traceless population plane and $\mathsf{C}_{\mathrm{coh}}$ is the off-diagonal coherence space. Let P_{pop} and $P_{\mathrm{coh}} = \mathbb{1} - P_{\mathrm{pop}}$ be the corresponding orthogonal projections. The adjoint action $\mathrm{Ad}_U(H) = UHU^\dagger$ has the block form

$$\mathrm{Ad}_U = \begin{pmatrix} M_U & B_U \\ G_U & C_U \end{pmatrix} := \begin{pmatrix} P_{\mathrm{pop}}\, \mathrm{Ad}_U\, P_{\mathrm{pop}} & P_{\mathrm{pop}}\, \mathrm{Ad}_U\, P_{\mathrm{coh}} \\ P_{\mathrm{coh}}\, \mathrm{Ad}_U\, P_{\mathrm{pop}} & P_{\mathrm{coh}}\, \mathrm{Ad}_U\, P_{\mathrm{coh}} \end{pmatrix}. \tag{71}$$

Thus $M_U : \mathsf{D} \to \mathsf{D}$, $G_U : \mathsf{D} \to \mathsf{C}_{\mathrm{coh}}$, $B_U : \mathsf{C}_{\mathrm{coh}} \to \mathsf{D}$, and $C_U : \mathsf{C}_{\mathrm{coh}} \to \mathsf{C}_{\mathrm{coh}}$. The first is the centered-population action of $\mathsf{B}(U)$; G_U generates coherence from a population contrast, B_U reads coherence back into a population contrast, and C_U transports coherence without reading it. The symbol $\star$ below denotes the adjoint of a block map with respect to the real Hilbert–Schmidt inner products. Orthogonality of the adjoint representation gives

$$B_U = G^\star_{U^\dagger}. \tag{72}$$

The subspaces $\operatorname{ran} G_U$ and $\operatorname{ran} G_{V^\dagger}$ are respectively the coherence frame created by U and the coherence frame readable by V.

Block multiplication isolates the entire qutrit composition defect:

$$M_{V \circ U} = M_V M_U + B_V G_U. \tag{73}$$

Theorem C.7 (Qutrit two-step criterion). *For arbitrary $U, V \in \mathrm{U}(3)$,*

$$\mathsf{B}(V \circ U) = \mathsf{B}(V) \circ \mathsf{B}(U) \iff B_V G_U = 0 \iff G^\star_{V^\dagger} G_U = 0. \tag{74}$$

Equivalently, the coherence frame created by U is orthogonal inside $\mathsf{C}_{\mathrm{coh}}$ to the coherence frame readable by V.

Proof. Both sides are bistochastic and therefore agree on the uniform population mode. Hence the Born kernels agree if and only if their actions agree on the centered population plane D. Equation (73) shows that their difference on this plane is $B_V G_U$. Equation (72) gives the final equivalence. □

The criterion can be written as four scalar equations. Put

$$h_x := P_x - \frac{\mathbb{1}_3}{3}, \qquad c_x(U) := G_U h_x, \qquad d_y(V) := G_{V^\dagger} h_y. \tag{75}$$

Since $\sum_x c_x(U) = \sum_y d_y(V) = 0$, two vectors from each triple span its frame. The entrywise defect is the cross-Gram matrix

$$\mathsf{B}(V \circ U)_{yx} - [\mathsf{B}(V) \circ \mathsf{B}(U)]_{yx} = \langle d_y(V), c_x(U) \rangle_{\mathrm{HS}}. \tag{76}$$

Indeed, expansion in the context projectors gives

$$\langle d_y(V), c_x(U) \rangle_{\mathrm{HS}} = |\langle y|V \circ U|x\rangle|^2 - \sum_z |\langle y|V|z\rangle|^2 |\langle z|U|x\rangle|^2, \tag{77}$$

which is exactly the displayed Born-kernel defect. Consequently Eq. (74) is equivalent to the vanishing of the 2×2 block with $x, y \in \{0, 1\}$. This is the qutrit replacement for the single qubit dot product in Eq. (62).

Definition C.8 (Qutrit coherence rank). The coherence rank of a qutrit unitary U is

$$r(U) := \operatorname{rank} G_U \in \{0, 1, 2\}. \tag{78}$$

The necessary-and-sufficient cross-Gram criterion above proves the qutrit claim used in the main text.

C.4 Prefixes and their faithful recurrence

We return to qubit steps and their scalar Born-kernel coordinate.

Definition C.9 (Prefix interference and prefix defect). For $k \geq 1$, put

$$\beta_k := \beta(U_k), \tag{79}$$

and

$$D_k^{\mathrm{pref}} := \beta(U_{k:1}) - \prod_{r=1}^{k} \beta_r. \tag{80}$$

For $k \geq 2$, define the one-step interference increment by

$$\varepsilon_k := \beta(U_{k:1}) - \beta(U_k)\beta(U_{k-1:1}). \tag{81}$$

The scalar ε_k is the interference created at the new prefix cut, while D_k^{pref} is the cumulative prefix-composition defect.

For $k \geq 2$,

$$D_k^{\mathrm{pref}} = \beta_k D_{k-1}^{\mathrm{pref}} + \varepsilon_k, \tag{82}$$

Indeed, substitute $\beta(U_{k:1}) = \beta_k \beta(U_{k-1:1}) + \varepsilon_k$ and add and subtract $\beta_k \prod_{r=1}^{k-1} \beta_r$. Iteration gives

$$D_L^{\mathrm{pref}} = \sum_{k=2}^{L} \varepsilon_k \prod_{m=k+1}^{L} \beta_m. \tag{83}$$

Prefix consistency at every step is equivalent to $\varepsilon_k = 0$ for every $k \geq 2$. By the pair classification,

$$\varepsilon_k = \chi(U_k, U_{k-1:1}) = 4 \operatorname{Re}\Big[a_0(U_k, U_{k-1:1}) \overline{a_1(U_k, U_{k-1:1})} \Big]. \tag{84}$$

Proposition C.10 (Infinite qubit sequences). *There exists an infinite sequence of genuinely mixing qubit steps for which every finite prefix satisfies the identity in Eq. (34).*

Proof. Choose a genuinely mixing first step. Suppose $U_1, \ldots, U_{k-1}$ have been chosen prefix-consistently, with every $|\beta_r| < 1$. Then

$$|\beta(U_{k-1:1})| = \left|\prod_{r=1}^{k-1} \beta_r\right| < 1, \tag{85}$$

so both amplitudes

$$\gamma_0 \coloneqq \langle 0|U_{k-1:1}|0\rangle, \qquad \gamma_1 \coloneqq \langle 1|U_{k-1:1}|0\rangle \tag{86}$$

are nonzero. Write $\gamma_j = |\gamma_j| e^{\imath\theta_j}$, choose any $\beta_k \in (-1, 1)$, and put

$$c_k \coloneqq \sqrt{\frac{1+\beta_k}{2}}, \qquad s_k \coloneqq \sqrt{\frac{1-\beta_k}{2}}. \tag{87}$$

Choose the first row of U_k to satisfy

$$\langle 0|U_k|0\rangle = c_k, \qquad \langle 0|U_k|1\rangle = \imath s_k e^{\imath(\theta_0-\theta_1)}, \tag{88}$$

and choose the second row as

$$\langle 1|U_k|0\rangle = \imath s_k e^{-\imath(\theta_0-\theta_1)}, \qquad \langle 1|U_k|1\rangle = c_k. \tag{89}$$

The resulting two rows are orthonormal. The two path amplitudes then have the common phase $e^{\imath\theta_0}$ and a relative factor of $\imath$. Hence $\chi(U_k, U_{k-1:1}) = 0$, while $\beta(U_k) = c_k^2 - s_k^2 = \beta_k$. The new prefix is consistent and $|\beta_k| < 1$ makes U_k genuinely mixing. Induction continues for all k; truncation gives a sequence of every prescribed finite length. □

Corollary C.11 (Constant-kernel infinite sequences). *Fix $-1 < \lambda < 1$, set $\theta = \arccos\lambda$, and define*

$$U_1 = R_{\sigma_y}(\theta), \qquad U_k = R_{\sigma_z}(\alpha_k) \circ R_{\sigma_x}(\theta), \qquad k \geq 2, \tag{90}$$

where $\alpha_k \in (-\pi/2, \pi/2)$ is determined by

$$\tan\alpha_k = \frac{\lambda^{k-1}\sqrt{1-\lambda^2}}{\sqrt{1-\lambda^{2(k-1)}}}. \tag{91}$$

Then every step has Born kernel $K(\lambda)$ and every finite prefix obeys

$$\mathsf{B}(U_\ell \circ \cdots \circ U_1) = K(\lambda^\ell) = \mathsf{B}(U_\ell) \circ \cdots \circ \mathsf{B}(U_1). \tag{92}$$

For every restart just before U_r, $r \geq 2$, however,

$$\mathsf{B}(U_{r+1} \circ U_r) = K\big(\lambda^2 - (1-\lambda^2)\cos\alpha_r\big) \neq K(\lambda^2) = \mathsf{B}(U_{r+1}) \circ \mathsf{B}(U_r). \tag{93}$$

Proof. Let $\boldsymbol{r}_k$ be the Bloch vector of $U_k \circ \cdots \circ U_1|0\rangle$. We claim

$$\boldsymbol{r}_k = \left(\sqrt{1-\lambda^{2k}}, 0, \lambda^k\right). \tag{94}$$

It holds at $k = 1$. If it holds at $k - 1$, then $R_{\sigma_x}(\theta)$ sends the last two components to

$$\left(-\lambda^{k-1}\sqrt{1-\lambda^2}, \lambda^k\right). \tag{95}$$

Equation (91) makes the following $R_{\sigma_z}(\alpha_k)$ rotation cancel the transverse y component and leaves Eq. (94). Therefore $\mathsf{B}(U_k) = K(\lambda)$ and the prefix kernel is $K(\lambda^k)$, proving Eq. (92) by $K(\lambda) \circ K(\mu) = K(\lambda \cdot \mu)$.

For a two-step segment beginning at U_r, direct Bloch rotation gives the polarization

$$\lambda^2 - (1 - \lambda^2)\cos\alpha_r. \tag{96}$$

Since $|\lambda| < 1$ and $\alpha_r \in (-\pi/2, \pi/2)$, this differs from λ^2, proving Eq. (93). At $\lambda = 1/2$, the local formulas in Eqs. (90) and (91) give the constant-amplitude specialization. At $\lambda = -1/2$, they give the signed companion, with

$$\tan\alpha_k = (-1)^{k-1}\frac{\sqrt{3}}{2\sqrt{4^{k-1} - 1}}. \tag{97}$$

□

C.5 Discrete path-integral formulation

Definition C.12 (Context paths, amplitudes, and weights). The composition law admits an exact sum-over-paths form. Fix

an entrance $x_0 = x$ and an endpoint $x_L = y$, and let

$$\Omega_{yx}^{(L)} := \{(x_0, \ldots, x_L) \in X^{L+1} : x_0 = x,\ x_L = y\} \tag{98}$$

be the context paths joining them. The amplitude and the corresponding stepwise path weight are

$$\mathcal{A}(\gamma) := \prod_{t=1}^{L}\langle x_t|U_t|x_{t-1}\rangle, \tag{99}$$

$$w(\gamma) := |\mathcal{A}(\gamma)|^2 = \prod_{t=1}^{L}\langle x_t|\mathsf{B}_{\mathcal{C}}(U_t)|x_{t-1}\rangle. \tag{100}$$

These expressions follow by inserting $\mathbb{1} = \sum_{x_t \in X}|x_t\rangle\langle x_t|$ at every intermediate boundary. Explicitly,

$$\langle y|U_{L:1}|x\rangle = \sum_{x_1,\ldots,x_{L-1}}\prod_{t=1}^{L}\langle x_t|U_t|x_{t-1}\rangle. \tag{101}$$

The endpoint-checked route adds amplitudes before applying the Born rule, whereas the stepwise-checked route adds their classical weights:

$$\mathsf{B}_{\mathcal{C}}(U_{L:1})_{yx} = \left| \sum_{\gamma \in \Omega_{yx}^{(L)}} \mathcal{A}(\gamma) \right|^2, \tag{102}$$

$$\big(\mathsf{B}_{\mathcal{C}}(U_L) \circ \cdots \circ \mathsf{B}_{\mathcal{C}}(U_1)\big)_{yx} = \sum_{\gamma \in \Omega_{yx}^{(L)}} w(\gamma). \tag{103}$$

Consequently,

$$\begin{aligned} &\mathsf{B}_{\mathcal{C}}(U_{L:1})_{yx} - \big(\mathsf{B}_{\mathcal{C}}(U_L) \circ \cdots \circ \mathsf{B}_{\mathcal{C}}(U_1)\big)_{yx} \\ &\quad = \sum_{\substack{\gamma, \gamma' \in \Omega_{yx}^{(L)} \\ \gamma \neq \gamma'}} \mathcal{A}(\gamma)\overline{\mathcal{A}}(\gamma') = 2\,\mathrm{Re} \sum_{\gamma < \gamma'} \mathcal{A}(\gamma)\overline{\mathcal{A}}(\gamma'). \end{aligned} \tag{104}$$

Here $\gamma < \gamma'$ denotes any fixed ordering of the finite path set; it merely counts each unordered pair once. Equation (104) is the path form of the Born–Chapman–Kolmogorov current. The quantum-to-stochastic composition square in Fig. 2 commutes exactly when this total cross-path contribution vanishes for every entrance x and endpoint y. This is an aggregate cancellation condition. Distinct paths may interfere, and the unitary steps may fail to commute. Chronology is retained by the time-labeled factors and their endpoints in Eq. (99), and by the ordered stochastic product in Eq. (103); scalar multiplication itself is commutative.

When the stepwise endpoint probability is nonzero, the stepwise path law can also be postselected on its endpoint:

$$\mathbb{P}_{\mathrm{step}}(\gamma \mid X_0 = x, X_L = y) = \frac{|\mathcal{A}(\gamma)|^2}{\sum_{\gamma' \in \Omega_{yx}^{(L)}} |\mathcal{A}(\gamma')|^2}. \tag{105}$$

This is the endpoint-conditioned bridge law of the dephased Markov chain.

Finally, beginning at an internal boundary replaces $\Omega_{yx}^{(L)}$ and its amplitudes by those of the prescribed suffix, with a freshly prepared $\mathcal{C}$-diagonal entrance state at that boundary. Cancellation of the cross terms for every prefix from the initial boundary therefore need not imply cancellation for the suffix. In path language, boundary instability is the failure of aggregate interference cancellation when the path sum is restarted at an internal boundary. Section C.7 contrasts this aggregate condition with pairwise consistency of fine-grained histories.

C.6 Exact relation to dephasing and Kolmogorov consistency

The vanishing-defect pair equation $\mathcal{J}_{\mathcal{C}}(V, U) = 0$ makes exact contact with work on dynamics that do not generate and subsequently detect coherence. Write $\mathcal{U}(\rho) = U\rho U^\dagger$ and $\mathcal{V}(\rho) = V\rho V^\dagger$. Then

$$\mathcal{M}_{\mathcal{C}} \circ \mathcal{V} \circ \mathcal{M}_{\mathcal{C}} \circ \mathcal{U} \circ \mathcal{M}_{\mathcal{C}} = \mathcal{M}_{\mathcal{C}} \circ \mathcal{V} \circ \mathcal{U} \circ \mathcal{M}_{\mathcal{C}} \iff \mathsf{B}_{\mathcal{C}}(V) \circ \mathsf{B}_{\mathcal{C}}(U) = \mathsf{B}_{\mathcal{C}}(V \circ U). \tag{106}$$

Indeed, after restricting their action to diagonal inputs and reading only diagonal outputs, the two superoperators have the displayed Born kernels as their matrices on the basis $\{P_x\}$. Our Born-kernel equation therefore recovers precisely the closed-unitary, fixed-context restriction of the non-coherence-generating-and-detecting (NCGD) condition [10, 11]. For one inserted measurement, this is no-signaling in time resolved by the entrance outcome, required for every context-basis preparation [6, 7, 12]. It does not assert that the intermediate quantum state is undisturbed.

NCGD already imposes the dephasing identity for all triples of times [11]; boundary stability specializes that requirement to a prescribed finite unitary sequence. Sakuldee, Taranto, and Milz give a pair of genuinely mixing qubit unitary steps satisfying the identity [13, Example 6]. Here we classify unitary solutions and distinguish an anchored family of composing prefixes from compatibility under independent preparation at internal boundaries.

We now prove the schedule-consistency equivalence used by the main text.

Definition C.13 (Checking schedules and schedule statistics). For every prescribed contiguous subpassage from boundary r to boundary s, where $0 \le r < s \le L$, a schedule is $\tau = (r = t_0 < t_1 < \cdots < t_m = s)$. Its sequential context statistics are

$$p_\tau(x_{t_1}, \ldots, x_{t_m} \mid x_r) := \prod_{j=1}^{m} \mathsf{B}_{\mathcal{C}}(U_{t_j:t_{j-1}+1})_{x_{t_j} x_{t_{j-1}}}. \tag{107}$$

The lower index $t_{j-1} + 1$ names the first step after boundary t_{j-1}; this shift is required because the t_j label boundaries whereas the subscripts of $U_{b:a}$ label the included steps. For fixed entrance outcome x_r, this product is a normalized joint Markov law obtained by reading $\mathcal{C}$ exactly at the scheduled boundaries. If the interior time t_j is deleted, Kolmogorov marginal consistency means

$$\sum_{x_{t_j}} p_\tau(x_{t_1}, \ldots, x_{t_m} \mid x_r) = p_{\tau \setminus \{t_j\}}(x_{t_1}, \ldots, \widehat{x}_{t_j}, \ldots, x_{t_m} \mid x_r), \tag{108}$$

where the hat denotes omission. Every internal-start test uses a fresh context atom, or by linearity an arbitrary $\mathcal{C}$-diagonal mixture, rather than the quantum state carried from an earlier passage. In the rhythmic representation of Section 5, the same schedule is written as the ordered composition $(t_1 - t_0, \ldots, t_m - t_{m-1})$ of $s - r$. With beat duration $\Delta > 0$, these differences specify successive sound holds. The schedule is *admissible* when its endpoint kernel equals $\mathsf{B}_{\mathcal{C}}(U_{s:r+1})$ for every entrance basis state, hence for every $\mathcal{C}$-diagonal mixture. This is a condition on the endpoint marginal; the stronger consistency under all readout insertions and deletions is stated in Proposition C.14.

Proposition C.14 (Boundary stability is full schedule consistency). *For a prescribed unitary sequence in a fixed atomic context, the following are equivalent:*

1. *the sequence is boundary-stable in the sense of Eq.* (35)*;*
2. *for every contiguous subpassage independently initialized in a $\mathcal{C}$-diagonal entrance state, its schedule statistics obey Kolmogorov marginal consistency under every insertion or deletion of a prescribed internal context readout.*

Proof. Fix a contiguous subpassage. Deleting one readout merges two adjacent unitary blocks. Boundary stability identifies the Born kernel of that merged block with the stochastic product of the two block kernels, so summing over the deleted outcome preserves the coarser distribution as in Eq. (108). Repetition proves consistency for every schedule and every diagonal entrance mixture. Conversely, compare on each subpassage the endpoint-checked schedule with the schedule containing every internal readout. Marginal consistency equates the direct interval Born kernel with the product of its one-step Born kernels, which is Eq. (35). □

Broader schedule-consistency settings are studied in [13, 14]. Proposition C.14 records the finite-sequence dictionary: boundary stability is fixed-context schedule consistency on every prescribed subpassage independently initialized in a $\mathcal{C}$-diagonal state, whereas prefix consistency retains only endpoint-versus-fully-refined comparisons from the initial boundary.

Boundary stability is operation-supplied divisibility across every prescribed cut. In stochastic divisibility, an intermediary need only exist [15]; here it must be the Born kernel of the actual unitary subpassage. The underlying closed-unitary processes remain operationally Markovian and completely positive divisible, even when these population kernels fail to compose [16, 17]. Quantum-channel divisibility is distinct: Wolf and Cirac ask whether a quantum channel factors through intermediate quantum channels [18].

C.7 Born-kernel endpoint composition does not imply consistent fine-grained histories

The consistent-histories framework assigns probabilities to a chosen family of multitime alternatives when its decoherence functional satisfies appropriate consistency conditions [3, 5, 49, 4]. Endpoint Born-kernel composition for a fixed passage is different from, and strictly weaker than, the following fixed-context fine-grained-history condition.

Definition C.15 (Fixed-context fine-grained-history consistency). Fix an entrance x, an endpoint y, and a sequence $U_1, \ldots, U_L$. For an intermediate context path $\alpha = (x_1, \ldots, x_{L-1})$, define

$$\mathcal{A}^{yx}_{\alpha} := \langle y|U_L P_{x_{L-1}} U_{L-1} \cdots P_{x_1} U_1|x\rangle, \qquad \mathfrak{D}^{yx}_{\alpha\beta} := \mathcal{A}^{yx}_{\alpha}\overline{\mathcal{A}^{yx}_{\beta}}. \tag{109}$$

Then

$$\begin{aligned}\mathsf{B}(U_L\circ\cdots\circ U_1)_{yx} &= \sum_{\alpha,\beta}\mathfrak{D}^{yx}_{\alpha\beta},\\ \big(\mathsf{B}(U_L)\circ\cdots\circ\mathsf{B}(U_1)\big)_{yx} &= \sum_{\alpha}\mathfrak{D}^{yx}_{\alpha\alpha}.\end{aligned}\tag{110}$$

Consequently, endpoint Born kernels compose precisely when

$$\sum_{\alpha\neq\beta}\mathfrak{D}^{yx}_{\alpha\beta}=0\qquad\text{for every } x,y.\tag{111}$$

$$\operatorname{Re}\mathfrak{D}^{yx}_{\alpha\beta}=0\qquad\text{for every } x,y \text{ and every } \alpha\neq\beta\tag{112}$$

is the fixed-context fine-grained specialization of weak decoherence. It is the pointwise weak-decoherence condition for this particular context-path family [3, 5, 49, 4].

The scalar $\mathfrak{D}^{yx}_{\alpha\beta}$ is the entrance–endpoint-resolved contribution to the standard decoherence functional. For the natural fine-grained context paths used below, medium decoherence requires $\mathfrak{D}^{yx}_{\alpha\beta}=0$ for every distinct pair, whereas weak decoherence requires $\operatorname{Re}\mathfrak{D}^{yx}_{\alpha\beta}=0$ pairwise. Either condition implies the aggregate cancellation in Eq. (111); the converse need not hold.

Condition	**What must vanish for each fixed x,y**	**Strength**
medium decoherence	every off-diagonal $\mathfrak{D}^{yx}_{\alpha\beta}$	strongest
weak decoherence	every $\operatorname{Re}\mathfrak{D}^{yx}_{\alpha\beta}$	intermediate
endpoint Born-kernel composition	only their aggregate sum	weakest

Proposition C.16 (Endpoint composition without weak decoherence). *For $\sigma\in\{\sigma_x,\sigma_y,\sigma_z\}$, let $R_\sigma(\theta)=e^{-\imath\theta\sigma/2}$. Choose $\varphi\in(0,\pi/2)$ with $\tan\varphi=2$, and set*

$$U_1=R_{\sigma_y}(\pi/3),\qquad U_2=R_{\sigma_x}(2\pi/3),\qquad U_3=R_{\sigma_y}(-\pi/3)\circ R_{\sigma_z}(-\varphi).\tag{113}$$

Both prefixes obey the fully refined Born-kernel identity:

$$\begin{aligned}\mathsf{B}(U_2\circ U_1)=\mathsf{B}(U_2)\circ\mathsf{B}(U_1)&=K(-1/4),\\ \mathsf{B}(U_3\circ U_2\circ U_1)=\mathsf{B}(U_3)\circ\mathsf{B}(U_2)\circ\mathsf{B}(U_1)&=K(-1/8).\end{aligned}\tag{114}$$

Nevertheless, the fine-grained histories of the complete three-step prefix are not weakly decoherent. Beginning instead at the internal boundary before U_2 destroys the aggregate cancellation:

$$\mathcal{J}_{\mathcal{C}}(U_3,U_2)=\frac{3\sqrt{5}}{20}(\sigma_x-\mathbb{1})\neq 0.\tag{115}$$

Thus the witness establishes endpoint composition for the complete prefix.

The sequence is not boundary-stable and therefore does not supply a genealogy on all contiguous intervals.

Proof. The individual kernels are $\mathsf{B}(U_1)=K(1/2)$, $\mathsf{B}(U_2)=K(-1/2)$, and $\mathsf{B}(U_3)=K(1/2)$. Direct multiplication gives Eq. (114). For the first two steps and $x=y=0$, the two path amplitudes satisfy

$$\mathfrak{D}^{00}_{01} = \frac{3\imath}{16}. \tag{116}$$

Thus medium decoherence already fails, although its real part vanishes.

For the complete prefix, order the four intermediate paths as $(0,0),(0,1),(1,0),(1,1)$. At $x = y = 0$, the real off-diagonal part of the decoherence matrix is

$$\operatorname{Re}\mathfrak{D}^{00}_{\mathrm{off}} = \frac{\sqrt{5}}{320}\begin{pmatrix} 0 & -18 & 0 & 3 \\ -18 & 0 & 9 & 0 \\ 0 & 9 & 0 & 6 \\ 3 & 0 & 6 & 0 \end{pmatrix}. \tag{117}$$

The matrix has nonzero entries, so pairwise weak decoherence fails, but the sum of all its entries is zero. The other three entrance–endpoint pairs have the same cancellation with the corresponding sign changes, which is equivalent to the second identity in Eq. (114). For the suffix, direct evaluation gives Eq. (115). □

The prefix composition identities in this witness therefore come from interference balance, not from absent interference. Cue instability is the loss of that balance when the remaining passage is restarted at an internal boundary.

D Quantitative bounds

We quantify schedule information and the composition current, then bound the number of full-support steps in a boundary-stable sequence.

D.1 Conditional information carried by the checking schedule

Definition D.1 (Checking-schedule information). Fix a contiguous passage containing steps a through b, where $1 \le a \le b \le L$, and define its coherent and fully checked endpoint kernels by

$$\mathsf{P}_{b:a} := \mathsf{B}_{\mathcal{C}}(U_{b:a}), \qquad \mathsf{Q}_{b:a} := \mathsf{B}_{\mathcal{C}}(U_b) \circ \cdots \circ \mathsf{B}_{\mathcal{C}}(U_a). \tag{118}$$

Let $X_{a-1} \sim \pi$ be a recorded entrance outcome and let an independent fair bit S select the protocol: $S = 0$ selects the coherent passage, and $S = 1$ selects the passage with unread context checks at every prescribed internal boundary. The final recorded outcome X_b then has joint law

$$\Pr(S = \sigma, X_{a-1} = x, X_b = y) = \frac{1}{2}\pi(x)(\mathsf{K}_\sigma)_{yx}, \qquad \mathsf{K}_0 = \mathsf{P}_{b:a}, \quad \mathsf{K}_1 = \mathsf{Q}_{b:a}. \tag{119}$$

For a discrete random variable A with law p_A, write $H_2(A) := -\sum_a p_A(a)\log_2 p_A(a)$, with $0\log_2 0 := 0$. For discrete (B,C), define $H_2(B \mid C) := \sum_c \Pr(C = c) H_2(B \mid C = c)$. The base-two conditional mutual information is

$$I(A; B \mid C) := H_2(B \mid C) - H_2(B \mid A, C).$$

It measures how much learning A reduces the uncertainty of B once C is known. We also write $I(A;B) := H_2(B) - H_2(B \mid A)$.

Equivalently, the base-two Shannon entropy of a probability vector p [19] is

$$H_2(p) := -\sum_y p_y \log_2 p_y, \qquad 0 \log_2 0 := 0. \tag{120}$$

Define the base-two Jensen–Shannon divergence [20] by

$$\mathrm{JS}_2(p,q) := H_2\left(\frac{p+q}{2}\right) - \frac{1}{2}H_2(p) - \frac{1}{2}H_2(q). \tag{121}$$

The *Born–Chapman–Kolmogorov schedule information* of the passage is

$$\mathcal{I}^{\mathrm{BCK}}_{\mathcal{C},\pi}(b:a) := I(S; X_b \mid X_{a-1}). \tag{122}$$

Here X_{a-1} is the boundary outcome immediately before the first included step U_a, while X_b is the outcome immediately after the last included step U_b.

Proposition D.2 (Conditional schedule information). *For the experiment in Eq. (119),*

$$\mathcal{I}^{\mathrm{BCK}}_{\mathcal{C},\pi}(b:a) = \sum_x \pi(x)\ \mathrm{JS}_2((\mathsf{P}_{b:a})_{\cdot x}, (\mathsf{Q}_{b:a})_{\cdot x}), \tag{123}$$

and $0 \le \mathcal{I}^{\mathrm{BCK}}_{\mathcal{C},\pi}(b:a) \le 1$ bit per run. If $\pi(x) > 0$ for every x, then

$$\mathcal{I}^{\mathrm{BCK}}_{\mathcal{C},\pi}(b:a) = 0 \quad \Longleftrightarrow \quad \mathsf{P}_{b:a} = \mathsf{Q}_{b:a}. \tag{124}$$

Consequently, for any fixed strictly positive π, a prescribed sequence is boundary-stable exactly when this conditional information vanishes on every contiguous passage. Prefix consistency requires the same condition only for passages with $a = 1$.

For the uniform entrance distribution,

$$\mathcal{I}^{\mathrm{BCK}}_{\mathcal{C},\mathrm{unif}}(b:a) \ge \frac{\|\mathsf{P}_{b:a} - \mathsf{Q}_{b:a}\|_{\mathrm{F}}^2}{8d\ln 2}. \tag{125}$$

For two steps, with $U_1 = U$ and $U_2 = V$, this becomes

$$\mathcal{I}^{\mathrm{BCK}}_{\mathcal{C},\mathrm{unif}}(2:1) \ge \frac{d-1}{8d\ln 2}\nu_{\mathcal{C}}(V,U)^2. \tag{126}$$

Proof. Conditioned on $X_{a-1} = x$ but not on S, the endpoint law is $m_x = [(\mathsf{P}_{b:a})_{\cdot x} + (\mathsf{Q}_{b:a})_{\cdot x}]/2$. Expanding $I(S; X_b \mid X_{a-1}) = H_2(X_b \mid X_{a-1}) - H_2(X_b \mid S, X_{a-1})$ gives Eq. (123). Strict concavity of Shannon entropy makes each Jensen–Shannon term nonnegative and zero exactly when its two arguments agree. Strict positivity of π therefore gives Eq. (124), whose all-interval version is Eq. (35). Finally, Pinsker's inequality gives

$$\mathrm{JS}_2(p,q) \ge \frac{\|p-q\|_1^2}{8\ln 2} \ge \frac{\|p-q\|_2^2}{8\ln 2}. \tag{127}$$

Averaging uniformly over the columns proves Eq. (125); the two-step specialization uses Eqs. (37) and (39). □

The conditioning on the entrance record is essential. If a uniform entrance is sampled and then discarded, bistochasticity makes both endpoint marginals uniform, so $I(S;X_b)=0$ can hold even when $\mathsf{P}_{b:a}\neq\mathsf{Q}_{b:a}$. This is classical Shannon information of the recorded variables (S,X_{a-1},X_b), not an entropy of the signed current, a count of hidden physical bits, or a generic measure of quantum non-Markovianity.

Coherence-mediated information return under classical reduction is established [21]; coherent-versus-dephased population discrepancies have also been quantified by trace distance [11]. The quantity above gives the conditional information about the chosen checking schedule.

The fair bit fixes the ordinary Jensen–Shannon normalization. More generally, if $\Pr(S=1)=\alpha\in(0,1)$, then the same proof replaces Eq. (121) by the weighted divergence

$$\mathrm{JS}_{2,\alpha}(p,q):=H_2((1-\alpha)p+\alpha q)-(1-\alpha)H_2(p)-\alpha H_2(q), \tag{128}$$

without changing the zero criterion. This biased form applies when a quantum computation supplies the schedule bit in Section F.3.

We therefore retain the schedule information in its native unit of bits and omit the optional qubit reference normalization.

Retaining the final quantum system R before the context measurement gives

$$I_c:=I(S;X_b\mid X_{a-1}),\qquad I_q:=I(S;R\mid X_{a-1}),$$

where I_q uses von Neumann entropies with base-two logarithms. For the independent fair checking bit S,

$$\boxed{0\le I_c\le I_q\le 1\text{ bit.}}$$

These inequalities follow from nonnegativity, data processing under the final measurement, and the entropy bound for a classical bit, respectively.

For the pair in Eq. (13), conditioned on either entrance label, the coherent density operator is pure, the checked density operator has eigenvalues $3/4,1/4$, and their equally weighted average has eigenvalues $(4\pm\sqrt{7})/8$. Hence

$$I_q=h_2\left(\frac{4+\sqrt{7}}{8}\right)-\frac{1}{2}h_2\left(\frac{3}{4}\right)\simeq 0.2504\text{ bits},$$

where $h_2(t)=-t\log_2 t-(1-t)\log_2(1-t)$. The common final unitary V leaves I_q unchanged.

D.2 A sharp Frobenius bound

Definition D.3 (Uniform mode and complex Hadamard unitaries). Here, set $d=|X|=|\mathcal{C}|=\dim\mathsf{H}$. Define the normalized equal-amplitude context ket

$$|{+_d}\rangle:=\frac{1}{\sqrt{d}}\sum_{x\in X}|x\rangle,\qquad J_d:=d|{+_d}\rangle\langle{+_d}|,$$

so that J_d is the $d \times d$ all-ones matrix. The normalized Frobenius current introduced in Eq. (39) has the sharp bound

$$0 \le \nu_{\mathcal{C}}(V, U) \le 1. \tag{129}$$

It measures the aggregate signed difference of the endpoint-checked and stepwise-checked kernels. Its largest possible value is attained.

A unitary $F \in \mathrm{U}(d)$ is a *complex Hadamard unitary* when every entry has modulus $1/\sqrt{d}$. Equivalently, its Born kernel is the uniform kernel J_d/d. With outcomes labeled $0, \ldots, d-1$, the Fourier matrix

$$F_d := \frac{1}{\sqrt{d}} \sum_{x,y=0}^{d-1} e^{2\pi\imath xy/d} |x\rangle\langle y|$$

is an example in every dimension.

Theorem D.4 (Sharp Born-composition bound). *For every $d \ge 2$,*

$$\sup_{U,V \in \mathrm{U}(d)} \nu_{\mathcal{C}}(V, U) = 1. \tag{130}$$

Every inverse complex-Hadamard pair $U = F$, $V = F^\dagger$ attains the bound.

Proof of Theorem D.4. Equation (53) gives

$$\mathcal{J}_{\mathcal{C}}(V, U) = \iota_{\mathcal{C}}^\dagger L_V Q_{\mathcal{C}} L_U \iota_{\mathcal{C}}. \tag{131}$$

Here $\iota_{\mathcal{C}}$ is an isometry, L_U and L_V are Hilbert–Schmidt unitaries, and $Q_{\mathcal{C}}$ is an orthogonal projection.

By Remark A.1, these contractions have operator norm one, and hence

$$\|\mathcal{J}_{\mathcal{C}}(V, U)\|_{\mathrm{op}} \le 1. \tag{132}$$

Both K_{end} and K_{step} in Eq. (36) are bistochastic. Hence the normalized equal-amplitude ket and its dual bra are respectively right and left null vectors of their difference:

$$\mathcal{J}_{\mathcal{C}}(V, U)|{+_d}\rangle = 0, \qquad \langle {+_d}|\mathcal{J}_{\mathcal{C}}(V, U) = 0. \tag{133}$$

Consequently $\operatorname{rank} \mathcal{J}_{\mathcal{C}}(V, U) \le d - 1$. Summing the squares of the singular values gives the standard rank–norm inequality

$$\|\mathcal{J}_{\mathcal{C}}(V, U)\|_{\mathrm{F}}^2 \le \operatorname{rank} \mathcal{J}_{\mathcal{C}}(V, U)\, \|\mathcal{J}_{\mathcal{C}}(V, U)\|_{\mathrm{op}}^2 \le d - 1. \tag{134}$$

This proves the upper bound.

Let F be complex Hadamard and take $U = F$, $V = F^\dagger$. Then

$$\mathsf{B}(V \circ U) = \mathbb{1}, \qquad \mathsf{B}(U) = \mathsf{B}(V) = \frac{J_d}{d}, \qquad \mathcal{J}_{\mathcal{C}}(V, U) = \mathbb{1} - \frac{1}{d} J_d. \tag{135}$$

The final matrix is the orthogonal projection onto the $(d-1)$-dimensional subspace orthogonal to $|{+_d}\rangle$. Its Frobenius norm is therefore $\sqrt{d-1}$, so $\nu_{\mathcal{C}}(F^\dagger, F) = 1$. $\square$

D.3 Why a boundary-stable qubit sequence has only two genuinely mixing steps

We use the defined qubit term "genuinely mixing" throughout.

Definition D.5 (Bloch rotation and cumulative axes). For a qubit unitary U, let $R_U \in \mathrm{SO}(3)$ be its Bloch-sphere rotation, defined by

$$U^\dagger(\boldsymbol{a}\cdot\boldsymbol{\sigma})U = (R_U^\mathsf{T}\boldsymbol{a})\cdot\boldsymbol{\sigma}, \qquad \boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z), \qquad \boldsymbol{z} = (0,0,1)^\mathsf{T}. \tag{136}$$

Taking the expectation in $|0\rangle$ gives

$$\beta(U) = \boldsymbol{z}\cdot R_U^\mathsf{T}\boldsymbol{z}, \tag{137}$$

because $\langle 0|\boldsymbol{a}\cdot\boldsymbol{\sigma}|0\rangle = \boldsymbol{z}\cdot\boldsymbol{a}$ for every $\boldsymbol{a}\in\mathbb{R}^3$. Define the cumulative Bloch axes

$$\boldsymbol{b}_0 \coloneqq \boldsymbol{z}, \qquad \boldsymbol{b}_k \coloneqq R_{U_{k:1}}^\mathsf{T}\boldsymbol{z}\in\mathbb{R}^3, \quad 1\le k\le L. \tag{138}$$

Thus $\boldsymbol{b}_0$ is the initial context axis, and $\boldsymbol{b}_k$ is that axis pulled back through the first k steps. For $1\le a\le b\le L$,

$$\beta(U_{b:a}) = \boldsymbol{b}_b\cdot\boldsymbol{b}_{a-1}, \tag{139}$$

and

$$\beta_k = \boldsymbol{b}_k\cdot\boldsymbol{b}_{k-1}. \tag{140}$$

Boundary stability is therefore equivalent to the Gram conditions

$$\boldsymbol{b}_b\cdot\boldsymbol{b}_{a-1} = \prod_{j=a}^{b}\beta_j \qquad (1\le a\le b\le L). \tag{141}$$

Definition D.6 (Qubit frame innovation). For $1\le k\le L$, the qubit frame innovation at step k is

$$\boldsymbol{e}_k \coloneqq \boldsymbol{b}_k - \beta_k\boldsymbol{b}_{k-1}. \tag{142}$$

It is the component of the new cumulative Bloch axis not predicted by the immediately preceding axis and its one-step polarization.

Theorem D.7 (Qubit boundary-stability bound). *Every boundary-stable qubit sequence contains at most two genuinely mixing steps.*

Proof. For the innovations in Definition D.6,

$$\boldsymbol{e}_k = \boldsymbol{b}_k - \beta_k\boldsymbol{b}_{k-1}. \tag{143}$$

For every $0\le r\le k-1$, Eq. (141) gives

$$\begin{aligned}\boldsymbol{e}_k\cdot\boldsymbol{b}_r &= \boldsymbol{b}_k\cdot\boldsymbol{b}_r - \beta_k\boldsymbol{b}_{k-1}\cdot\boldsymbol{b}_r & (144)\\ &= 0. & (145)\end{aligned}$$

Also

$$\|\boldsymbol{e}_k\|^2 = 1-\beta_k^2. \tag{146}$$

Thus every genuinely mixing step, characterized by $|\beta_k| < 1$, adds a new nonzero direction orthogonal to the span of all previous $\boldsymbol{b}_r$. Starting from the one-dimensional span of $\boldsymbol{b}_0$ and remaining inside $\mathbb{R}^3$, there is room for at most two such innovations. Hence at most two steps can genuinely mix. □

Equivalently, the Gram matrix in Eq. (141) is the correlation matrix of the signed two-state population mode and has rank

$$1 + |\{k : |\beta_k| < 1\}|, \tag{147}$$

which cannot exceed three for Bloch vectors.

D.4 Full-support capacity and prime-dimensional saturation

This subsection proves the full-support step-count bound; primality is used only to attain it. By Definition A.8, a step U_k has full support precisely when

$$\mathsf{B}(U_k)_{yx} > 0 \qquad \text{for every } x, y. \tag{148}$$

For qubits, full support is exactly the defined genuinely mixing condition. The upper bound holds in every finite dimension; primality enters only in the sharp construction below.

D.4.1 Context frames and dimension accounting

The manuscript-specific innovation terminology is introduced only in numbered Definition D.10.

Definition D.8 (Population-deviation space and context frames). Let

$$\mathcal{R}_d^0 := \left\{ v \in \mathbb{R}^d : \sum_x v_x = 0 \right\} \tag{149}$$

be the $(d-1)$-dimensional space of population deviations from the uniform distribution.

Embed it isometrically in the real Hilbert space $\mathsf{Herm}_0(d)$ of traceless Hermitian operators by

$$\mathrm{Diag}_{\mathcal{C}}(v) := \sum_x v_x |x\rangle\langle x|. \tag{150}$$

Here $|x\rangle$ denotes the xth fixed context-basis vector, so $|x\rangle\langle x|$ is the matrix unit with its only nonzero entry in position (x, x); thus $\mathrm{Diag}_{\mathcal{C}}(v)$ is simply the usual diagonal matrix with diagonal v. The inner products are Euclidean and Hilbert–Schmidt, respectively, and

$$\dim_{\mathbb{R}} \mathsf{Herm}_0(d) = d^2 - 1. \tag{151}$$

Define the context-frame isometries of the cumulative passages by

$$\begin{aligned}
&F_k : \mathcal{R}_d^0 \longrightarrow \mathsf{Herm}_0(d), \qquad 0 \le k \le L,\\
&F_0(v) := \mathrm{Diag}_{\mathcal{C}}(v),\\
&F_k(v) := U_{k:1}^{\dagger}\,\mathrm{Diag}_{\mathcal{C}}(v) U_{k:1}, \qquad 1 \le k \le L.
\end{aligned} \tag{152}$$

The map $\mathrm{Diag}_{\mathcal{C}}$ is an isometry and unitary conjugation preserves the Hilbert–Schmidt inner product, so F_k identifies a zero-sum population deviation with the corresponding fixed-context diagonal observable pulled back through the first k steps.

For $1 \le a \le b \le L$, direct evaluation of the Hilbert–Schmidt inner product gives, for $v, w \in \mathcal{R}_d^0$,

$$\begin{aligned}
&\langle F_b(v), F_{a-1}(w)\rangle_{\mathrm{HS}} = \sum_{x,y} v_x w_y \,|\langle x|U_{b:a}|y\rangle|^2\\
&= \langle v, \mathsf{B}(U_{b:a}) w\rangle_2.
\end{aligned} \tag{153}$$

Equivalently,

$$F_b^* \circ F_{a-1} = \mathsf{B}(U_{b:a})|_{\mathcal{R}_d^0}. \tag{154}$$

Definition D.9 (One-step frame transition). Here F_b^* is the adjoint between real inner-product spaces. Put

$$T_k := F_k^* \circ F_{k-1} = \mathsf{B}(U_k)|_{\mathcal{R}_d^0}. \tag{155}$$

Boundary stability preserves the zero-sum population subspace because every Born kernel is bistochastic. Thus T_k is the signed-population restriction of a stochastic kernel; it is a linear contraction on $\mathcal{R}_d^0$, not itself a probability transition on a simplex.

Boundary stability is therefore precisely the block Gram rule

$$F_b^* \circ F_{a-1} = T_b \circ T_{b-1} \circ \cdots \circ T_a \qquad (1 \le a \le b \le L). \tag{156}$$

Definition D.10 (Context-frame innovation). The *context-frame innovation* is the part of the next frame not predicted by its immediate predecessor:

$$E_k := F_k - F_{k-1} \circ T_k^*. \tag{157}$$

Thus $E_k(v)$ is the residual diagonal observable after the preceding context frame has accounted for the linear prediction $T_k^* v$. Its *innovation rank* is

$$r_k^{\mathrm{inn}} := \mathrm{rank}(E_k). \tag{158}$$

For every $0 \le r \le k-1$, Eq. (156) yields

$$\begin{aligned}
&E_k^* \circ F_r = F_k^* \circ F_r - T_k \circ F_{k-1}^* \circ F_r\\
&\qquad = T_k \circ T_{k-1} \circ \cdots \circ T_{r+1} - T_k \circ T_{k-1} \circ \cdots \circ T_{r+1} = 0.
\end{aligned} \tag{159}$$

Thus every innovation is orthogonal to all earlier context frames. Since

$$\mathrm{ran}\, E_j \subseteq \mathrm{ran}\, F_j + \mathrm{ran}\, F_{j-1},$$

the ranges of $E_1, \dots, E_L$ are mutually orthogonal and are orthogonal to $\operatorname{ran} F_0$. Expanding the definition and using

$$F_k^* F_k = F_{k-1}^* F_{k-1} = \mathbb{1},$$
$$F_k^* F_{k-1} = T_k,$$

gives

$$E_k^* \circ E_k = \mathbb{1} - T_k \circ T_k^*. \tag{160}$$

Every bistochastic matrix is a Euclidean contraction by Eq. (41). Hence $\mathbb{1} - T_k \circ T_k^*$ is positive semidefinite. Equation (160) gives

$$r_k^{\mathrm{inn}} = \operatorname{rank}(\mathbb{1} - T_k \circ T_k^*) = \operatorname{rank}(E_k). \tag{161}$$

The original frame $\operatorname{ran} F_0$ consumes $d-1$ dimensions of $\mathsf{Herm}_0(d)$, and its mutually orthogonal innovations consume $\sum_k r_k^{\mathrm{inn}}$ more. Therefore

$$(d-1) + \sum_{k=1}^{L} r_k^{\mathrm{inn}} \le d^2 - 1, \tag{162}$$

and hence

$$\sum_{k=1}^{L} r_k^{\mathrm{inn}} \le d(d-1). \tag{163}$$

The same innovation count has an information-theoretic form.

Definition D.11 (Mean conditional linear entropy). For a bistochastic kernel K, define its mean conditional linear entropy by

$$\ell_2(K) := \frac{1}{d} \sum_x \left(1 - \sum_y K_{yx}^2\right) = 1 - \frac{\|K\|_{\mathrm{F}}^2}{d}. \tag{164}$$

This is a linear, or Tsallis–2, entropy. Equivalently, it is one minus the collision probability, averaged over uniformly sampled entrances. It is dimensionless.

Proposition D.12 (Finite linear-entropy budget). *Every boundary-stable d-dimensional sequence obeys*

$$\sum_{k=1}^{L} \ell_2(\mathsf{B}(U_k)) \le d-1. \tag{165}$$

The prime-dimensional construction in Theorem D.13 attains equality.

Proof. The uniform mode contributes one to $\|\mathsf{B}(U_k)\|_{\mathrm{F}}^2$, and its restriction to the orthogonal zero-sum sector is T_k. Therefore Eqs. (160) and (164) give

$$d\,\ell_2(\mathsf{B}(U_k)) = \operatorname{Tr}(\mathbb{1} - T_k T_k^*) = \|E_k\|_{\mathrm{F}}^2 \le r_k^{\mathrm{inn}}. \tag{166}$$

The last inequality holds because the positive contraction $\mathbb{1} - T_k T_k^*$ has eigenvalues in $[0,1]$. Summing and using Eq. (163) proves the bound. In the prime construction, every one-step kernel is J_d/d, so every term equals $1 - 1/d$ and the d terms sum to $d-1$. □

If $K = \mathsf{B}(U_k)$ has full support, then KK^{T} is a strictly positive bistochastic matrix. The Perron–Frobenius theorem makes its eigenvalue 1 simple, with the uniform vector as its eigendirection. The orthogonal complement of that vector is precisely $\mathcal{R}_d^0$, so every singular value of $T_k = K|_{\mathcal{R}_d^0}$ is strictly smaller than one. It follows that $\mathbb{1} - T_k \circ T_k^*$ is positive definite on $\mathcal{R}_d^0$ and

$$r_k^{\mathrm{inn}} = d - 1. \tag{167}$$

Each full-support step therefore consumes $d-1$ dimensions of the budget in Eq. (163); at most d such steps can occur.

D.4.2 Sharp full-support capacity theorem

Theorem D.13 (Sharp full-support capacity). *Let $d \geq 2$ and let $U_1, \ldots, U_L$ be unitary operations on $\mathbb{C}^d$. Suppose every contiguous interval obeys the Born-kernel composition law:*

$$\begin{aligned} \mathsf{B}(U_{b:a}) = \mathsf{B}(U_b) \circ \cdots \circ \mathsf{B}(U_a), \\ (1 \leq a \leq b \leq L). \end{aligned} \tag{168}$$

Then at most d one-step kernels $\mathsf{B}(U_k)$ have full support. For every prime d, this bound is attained by a boundary-stable sequence of exactly d full-support steps, for which every nonempty interval satisfies

$$\begin{aligned} \mathsf{B}(U_{b:a}) &= \mathsf{B}(U_b) \circ \cdots \circ \mathsf{B}(U_a) \\ &= |+_d\rangle\langle +_d|, \\ &\quad (1 \leq a \leq b \leq d). \end{aligned} \tag{169}$$

where $|+_d\rangle$ is the normalized equal-amplitude context ket.

Here d is the Hilbert-space dimension, whereas L is the independent number of prescribed steps. In the saturating construction $L = d$, so its d steps have $d+1$ boundary basis frames.

Proof. If m one-step kernels have full support, then $m(d-1) \leq \sum_k r_k^{\mathrm{inn}} \leq d(d-1)$, hence $m \leq d$.

For prime d, to attain the bound, choose a complete family of $d+1$ mutually unbiased bases [51], beginning with the computational basis. Write $V_1, V_2, \ldots, V_{d+1}$ for their unitary basis maps, where $V_a|x\rangle$ is the xth vector of basis a and $V_1 = \mathbb{1}$. Thus, whenever $a \neq b$,

$$\left|\langle x|V_a^\dagger V_b|y\rangle\right|^2 = \frac{1}{d} \qquad \text{for every } x, y. \tag{170}$$

Use these bases as cumulative basis frames:

$$\begin{aligned} U_k &:= V_{k+1}^\dagger \circ V_k, \\ U_{k:1} &= V_{k+1}^\dagger, \qquad 1 \leq k \leq d. \end{aligned} \tag{171}$$

For every nonempty contiguous interval, the intermediate factors telescope,

$$U_{b:a} = V_{b+1}^\dagger \circ V_a, \qquad 1 \leq a \leq b \leq d, \tag{172}$$

and the distinct endpoint bases are mutually unbiased. Therefore

$$\mathsf{B}(U_{b:a}) = |+_d\rangle\langle +_d|. \tag{173}$$

Every one-step kernel is the same projector, which is idempotent:

$$\big(|+_d\rangle\langle +_d|\big)^m = |+_d\rangle\langle +_d| \qquad (m \geq 1). \tag{174}$$

The Born composition law consequently holds on every interval. All d steps have full support, so the upper bound is attained. □

For completeness, when d is an odd prime, let $\mathbb{F}_d$ be the finite field with d elements and write $\omega = e^{2\pi\imath/d}$. Besides the computational basis, one may take the d bases

$$|a,b\rangle := \frac{1}{\sqrt{d}} \sum_{x\in\mathbb{F}_d} \omega^{ax^2+bx}|x\rangle, \qquad a,b \in \mathbb{F}_d. \tag{175}$$

For fixed a these vectors are orthonormal, while distinct values of a give quadratic Gauss sums of magnitude $\sqrt{d}$. For $d = 2$, the eigenbases of σ_z, σ_x, and σ_y supply the three mutually unbiased bases and hence two steps.

E The analytic Walsh three-step circuit

This section gives the full verification of the analytic circuit and the composition identities stated locally in Theorem E.3. For self-contained checking, define

$$H = \frac{\sigma_x + \sigma_z}{\sqrt{2}}, \qquad W = H \otimes H, \qquad \Phi(a,b,c) = \mathrm{diag}\big(1, e^{\imath\pi a}, e^{\imath\pi b}, e^{\imath\pi c}\big), \tag{176}$$

and let C_{NOT} be CNOT with the first qubit controlling the second. For real phase parameters ξ, η, ζ, put

$$\begin{aligned}
&Q_1 = W \circ \Phi(\xi, 0, \xi) \circ W \circ \Phi(1/2, 0, 1/2) \circ W,\\
&Q_2 = W \circ \Phi(-1/2, 1/2, -1) \circ W \circ \Phi(1/2 - \xi, 1/2, -\xi) \circ W,\\
&Q_3 = W \circ \Phi(1/2, 0, 1/2) \circ W \circ \Phi(\eta, \zeta, \zeta - \eta) \circ W,\\
&U_1 = C_{\mathrm{NOT}} \circ Q_1, \qquad U_2 = C_{\mathrm{NOT}} \circ Q_2 \circ C_{\mathrm{NOT}}, \qquad U_3 = C_{\mathrm{NOT}} \circ Q_3 \circ C_{\mathrm{NOT}}.
\end{aligned} \tag{177}$$

These definitions make the construction self-contained.

This all-interval family supplies the cue-stable counterpart to the schedule-selective audible construction in Section 5. Here W, without a subscript, is the normalized Walsh transform and satisfies $W^{\mathsf{T}} = W = W^{-1}$. At each boundary the final CNOT of one step is adjacent to the initial CNOT of the next and the pair squares to the identity. The gates are retained because the step boundaries and every contiguous subinterval are part of the claim.

Figure 5 expands the three steps at gate level and compares the endpoint-only and stepwise-checking protocols.

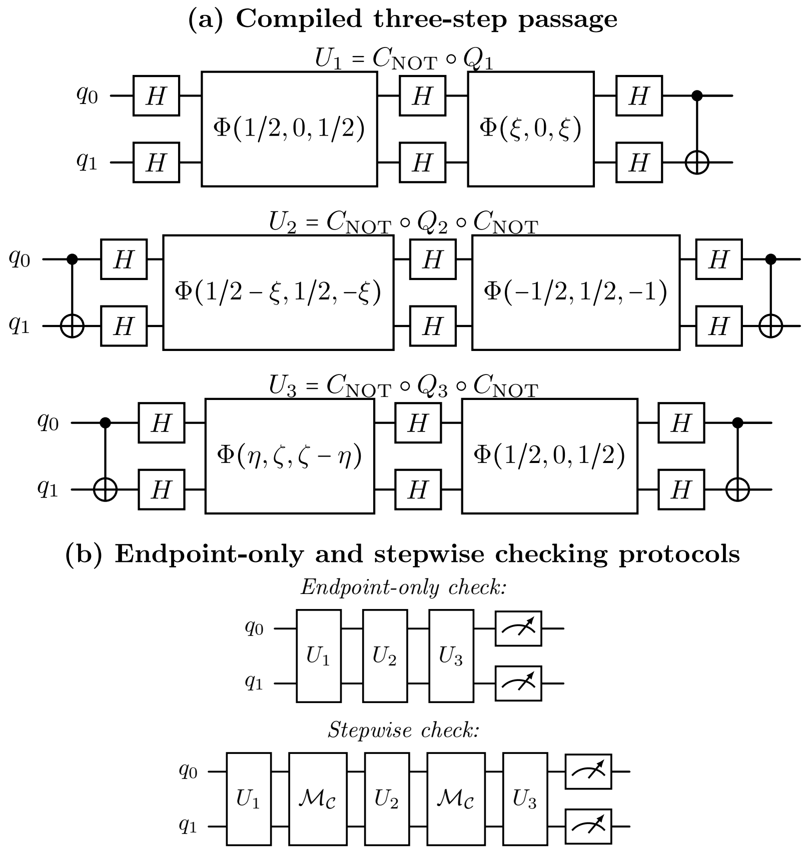

Figure 5: **Two-qubit circuits for the analytic Walsh passage.** Panel (a) compiles the three steps in Eq. (177); circuit time runs from left to right. Panel (b) compares the same passage under an endpoint-only readout and unread context checks after the first two steps. The next figure gives the record-register realization of each $\mathcal{M}_\mathcal{C}$ block. Theorem E.3 proves equality of the two endpoint laws on every contiguous interval; the dense specialization retains nonzero pairwise path interference.

Fixed-context measurement key.

The quantum action of each $\mathcal{M}_\mathcal{C}$ block has the following record-register realization.

$$\rho \longmapsto \mathcal{M}_\mathcal{C}(\rho) \coloneqq \sum_{x \in \{0,1\}^2} |x\rangle\langle x|\, \rho\, |x\rangle\langle x|.$$

Figure 6: **Ancilla implementation of the fixed-context check $\mathcal{M}_\mathcal{C}$.** Here $\mathcal{C}$ is the computational context. The CNOTs copy its two binary labels into fresh record targets, which are measured in the Z basis; double wires denote classical records. Retaining a record makes its outcome available for sonification; marginalizing it implements the non-selective channel $\mathcal{M}_\mathcal{C}$.

E.1 Walsh-coordinate representation

The coined label is isolated for author review after the coordinate definition below.

Definition E.1 (Walsh-coordinate representation). For a classical 4×4 matrix K, define $\widetilde{K} := WKW$. The tilde denotes the same linear map in Walsh coordinates; in particular, $\widetilde{\mathsf{B}(U)} = W\mathsf{B}(U)W$. We call this coordinate matrix its *Walsh shadow*; it need not be stochastic or entrywise nonnegative. The zeroth Walsh coordinate is the constant mode and the other three coordinates are the nonconstant binary parity modes. If E_{ab} is the matrix unit with a single 1 in row a, column b, then

$$E_{ab}E_{cd} = \delta_{bc}E_{ad}. \tag{178}$$

Lemma E.2 (Walsh-coordinate kernels). *For the three-parameter family in Eqs.* (176)–(177)*,*

$$\begin{aligned}\widetilde{\mathsf{B}(U_1)} &= E_{00} + \sin(\pi\xi)E_{31},\\ \widetilde{\mathsf{B}(U_2)} &= E_{00} + \sin(\pi\xi)E_{31},\\ \widetilde{\mathsf{B}(U_3)} &= E_{00} + \sin(\pi\eta)E_{13},\end{aligned} \tag{179}$$

and the coherently composed segments satisfy

$$\begin{aligned}\mathsf{B}(\widetilde{U_2 \circ U_1}) &= E_{00},\\ \mathsf{B}(\widetilde{U_3 \circ U_2}) &= E_{00} + \sin(\pi\xi)\sin(\pi\eta)E_{11},\\ \mathsf{B}(U_3 \circ \widetilde{U_2} \circ U_1) &= E_{00}.\end{aligned} \tag{180}$$

Proof. Put $x = e^{\imath\pi\xi}$, $y = e^{\imath\pi\eta}$, and $z = e^{\imath\pi\zeta}$. Substitution of the six diagonal phase layers, multiplication of the 4×4 matrices, entrywise squared moduli, and conjugation by W leave only the entries in Table 1.

Table 1: Nonzero entries of the Walsh shadows.

Segment	Constant entry	Additional entry
U_1	$(0,0) = 1$	$(3,1) = \imath(1-x^2)/(2x) = \sin(\pi\xi)$
U_2	$(0,0) = 1$	$(3,1) = \imath(1-x^2)/(2x) = \sin(\pi\xi)$
U_3	$(0,0) = 1$	$(1,3) = \imath(1-y^2)/(2y) = \sin(\pi\eta)$
$U_2 \circ U_1$	$(0,0) = 1$	none
$U_3 \circ U_2$	$(0,0) = 1$	$(1,1) = \sin(\pi\xi)\sin(\pi\eta)$
$U_3 \circ U_2 \circ U_1$	$(0,0) = 1$	none

The parameter z, hence ζ, cancels from all six Born shadows. The calculation is symbolic and holds for arbitrary real ξ, η, ζ. $\square$

Theorem E.3 (Analytic three-step composition). *For the real-parameter family in*

Eqs. (176)–(177), every nontrivial contiguous interval is operation-supplied compositional:

$$\begin{aligned}\mathsf{B}(U_2\circ U_1)&=\mathsf{B}(U_2)\circ\mathsf{B}(U_1),\\ \mathsf{B}(U_3\circ U_2)&=\mathsf{B}(U_3)\circ\mathsf{B}(U_2),\\ \mathsf{B}(U_3\circ U_2\circ U_1)&=\mathsf{B}(U_3)\circ\mathsf{B}(U_2)\circ\mathsf{B}(U_1).\end{aligned}\tag{181}$$

Hence the three-step passage is boundary-stable for every real ξ,η,ζ.

Proof of Theorem E.3. Put $a=\sin(\pi\xi)$ and $b=\sin(\pi\eta)$. Since $W^2=\mathbb{1}$, stochastic kernel composition may be checked in Walsh coordinates. By Eq. (178),

$$\begin{aligned}(E_{00}+aE_{31})^2&=E_{00},\\ (E_{00}+bE_{13})(E_{00}+aE_{31})&=E_{00}+abE_{11},\\ (E_{00}+abE_{11})(E_{00}+aE_{31})&=E_{00}.\end{aligned}\tag{182}$$

These are exactly the Walsh-coordinate matrices of the coherent endpoint Born kernels in Eq. (180); conjugating back by W proves every identity in Eq. (181). The density, interference, and irreducibility claims at the specialization are proved in the following subsections. □

E.2 Complete dense specialization

Set $\xi=\eta=1/6$ and $\zeta=0$. Define

$$K_{31}^{\mathrm{d}}:=W\left(E_{00}+\frac{1}{2}E_{31}\right)W,\qquad K_{13}^{\mathrm{d}}:=W\left(E_{00}+\frac{1}{2}E_{13}\right)W,\tag{183}$$

and

$$K_{\mathrm{unif}}:=WE_{00}W=|+_4\rangle\langle+_4|=\frac{1}{4}J_4,\qquad K_{11}^{\mathrm{d}}:=W\left(E_{00}+\frac{1}{4}E_{11}\right)W.\tag{184}$$

Here $|+_4\rangle$ is the $d=4$ case of the equal-amplitude ket defined in Section D.2. For direct numerical checking, the two one-step kernels are

$$K_{31}^{\mathrm{d}}=\frac{1}{8}\begin{pmatrix}3&1&3&1\\1&3&1&3\\1&3&1&3\\3&1&3&1\end{pmatrix},\qquad K_{13}^{\mathrm{d}}=\frac{1}{8}\begin{pmatrix}3&1&1&3\\1&3&3&1\\3&1&1&3\\1&3&3&1\end{pmatrix}.\tag{185}$$

The complete interval calculation is

$$\mathsf{B}(U_1)=K_{31}^{\mathrm{d}},\quad \mathsf{B}(U_2)=K_{31}^{\mathrm{d}},\quad \mathsf{B}(U_3)=K_{13}^{\mathrm{d}},\tag{186}$$

and

$$\begin{aligned}\mathsf{B}(U_2\circ U_1)&=K_{31}^{\mathrm{d}}\circ K_{31}^{\mathrm{d}}=K_{\mathrm{unif}},\\ \mathsf{B}(U_3\circ U_2)&=K_{13}^{\mathrm{d}}\circ K_{31}^{\mathrm{d}}=K_{11}^{\mathrm{d}},\\ \mathsf{B}(U_3\circ U_2\circ U_1)&=K_{13}^{\mathrm{d}}\circ K_{31}^{\mathrm{d}}\circ K_{31}^{\mathrm{d}}=K_{\mathrm{unif}}.\end{aligned}\tag{187}$$

These equations give the complete interval algebra of the three-step family.

E.3 Multipath interference

Consider the first two steps, input 0, and output 1. The four amplitude paths through the intermediate labels $m = 0, 1, 2, 3$ are $z_m := \langle 1|U_2|m\rangle\langle m|U_1|0\rangle$, namely

$$z_0 = \frac{\sqrt{3}}{8}, \qquad z_1 = -\frac{\sqrt{3}}{8}, \qquad z_2 = \frac{\imath}{8}, \qquad z_3 = \frac{3\imath}{8}. \tag{188}$$

Every path is nonzero, and

$$\left|\sum_{m=0}^{3} z_m\right|^2 = \left|\frac{\imath}{2}\right|^2 = \frac{1}{4} = \sum_{m=0}^{3} |z_m|^2. \tag{189}$$

The relation between the two probabilities is

$$\left|\sum_m z_m\right|^2 = \sum_m |z_m|^2 + 2 \sum_{0\le r<s\le 3} \operatorname{Re}(z_r \overline{z_s}). \tag{190}$$

Only two real cross terms are nonzero:

$$2\operatorname{Re}(z_0\overline{z_1}) = -\frac{3}{32}, \qquad 2\operatorname{Re}(z_2\overline{z_3}) = +\frac{3}{32}. \tag{191}$$

They cancel. The equality therefore comes from interference balance, not from absent or separated amplitude paths.

F Musical realizations and prediction complexity

We first give the two-pitch realization and the qutrit rhythm selector, including their sound-output conventions. We then prove the transition- prediction and sampling results. The prediction problem applies equally when the outputs label states of a finite state machine.

F.1 Qubit Ostinato: an audible realization

Example F.1 (Qubit Ostinato). The *Qubit Ostinato* is the audible $K(1/2)$ member of the exact constant-kernel family proved in Corollary C.11. It maps the computational basis to the adjacent pitch space

$$A_{\text{Ostinato}} = \{\mathrm{E}_2, \mathrm{F}_2\}, \qquad \ell_{\text{snd}}(|0\rangle) = \mathrm{E}_2, \qquad \ell_{\text{snd}}(|1\rangle) = \mathrm{F}_2.$$

Consequently,

$$\begin{aligned} \mathsf{B}(U_k) &= K(1/2) = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}, \\ \mathsf{B}(U_k \circ \cdots \circ U_1) &= K(2^{-k}). \end{aligned} \tag{192}$$

Under readout after every operation, the pitch changes with probability 1/4 at each step, while every finite prefix has the same endpoint distribution as the coherent passage. Figure 7 makes the first two updates explicit.

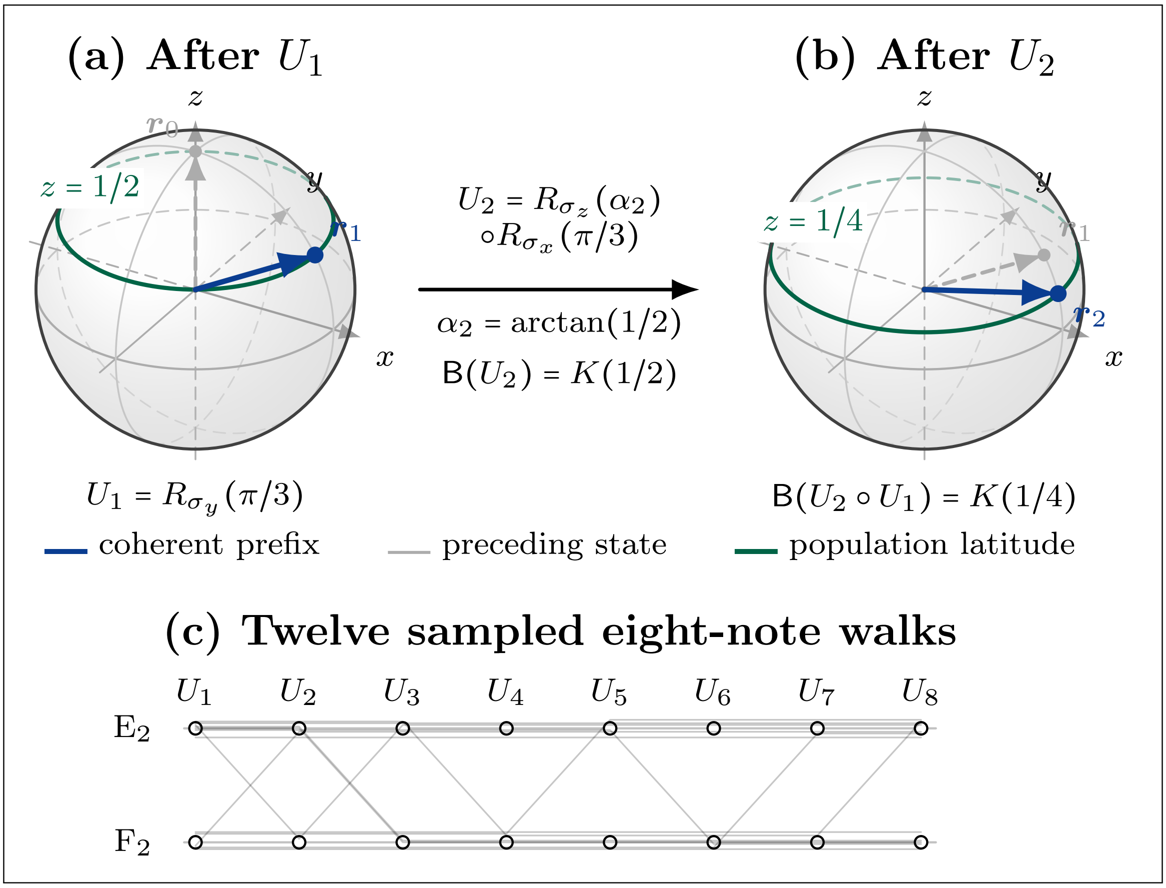

Figure 7: **Qubit Ostinato in the two-note space $\{\mathbf{E_2}, \mathbf{F_2}\}$.** Here Pr(change) = 1/4, with $|0\rangle \mapsto E_2$ and $|1\rangle \mapsto F_2$. Panels (a) and (b) are orthographic views of the Bloch sphere after the first two steps of the constant-kernel sequence in Eqs. (90) and (91). The coherent Bloch vectors are $\boldsymbol{r}_1 = (\sqrt{3}/2, 0, 1/2)$ and $\boldsymbol{r}_2 = (\sqrt{15}/4, 0, 1/4)$. Gray arrows show the preceding states, beginning at $\boldsymbol{r}_0 = (0, 0, 1)$. Green latitude circles mark the population polarizations $z = 1/2$ and $z = 1/4$; rear arcs are lighter and dashed. Panel (c) shows twelve reproducible eight-note samples of the associated two-state stochastic walk, beginning with the first audible outcome after U_1.

The Qubit Ostinato is an audible realization of the separation between initial- and internal-cue tests: composition from the initial cue does not by itself make the score cue-stable.

F.2 Qutrit rhythm selector

Let J_3 denote the 3×3 all-ones matrix. The qutrit selector used in Figure 4 is

$$P_3|s\rangle = |(s+1) \bmod 3\rangle, \qquad a = \frac{-1 + \imath\sqrt{55}}{8},$$
$$V^{\mathrm{sel}} = aP_3 + \frac{1}{4}(J_3 - P_3).$$

Its Born kernel is given in Eq. (30).

Score and sound output. A score prescribes an ordered sequence of unitary operations in a fixed context, together with a classical sound assignment; its readout schedule is specified separately. In the musical dictionary, each operation is a note and each temporal boundary is a cue. At an enabled boundary b, outcome x prepares the post-measurement state $|x\rangle$ and the classical map $\ell_{\mathrm{snd},b} : X \to \mathsf{A}_{\mathrm{snd}}$ assigns a sound. The boundary dependence of this map changes the voicing, while x retains its role in the subsequent quantum evolution.

The entrance sound is specified separately from the entrance quantum state and is held until the first scheduled readout. Every enabled readout starts its assigned sound anew, including when the pitch repeats; between readouts the sound is held. Thus the spacings between successive readouts determine the rhythm, with an externally chosen beat duration Δ.

Displayed performance. In the two-qubit realization, each data pass starts in $|00\rangle$ and the entrance sound carries over from the preceding endpoint. The selector starts in state $S = 0$ and thereafter is prepared in its preceding observed state before V^{sel} and readout. Its outcomes $0, 1, 2$ choose $(2, 1)$, (3), and $(1, 1, 1)$, respectively. The sound map is the boundary-dependent pitch table in Section 5.

The displayed seed-1766 window begins at an observed endpoint and contains five complete three-beat data passes, with selector outcomes $1, 2, 0, 1, 2$. The mandatory final readout is at $t = 15\Delta$. Its assigned pitch C_4 is sounded until $t = 16\Delta$, followed by one silent beat ending at $t = 17\Delta$. This terminal one-beat hold is a playback convention after the measured passage and adds no quantum operation or readout. The main-text circuit in Eq. (31) and Figure 4 give the complete realization and the 256 eight-readout comparison excerpts. Their sampled outcomes and durations are independent of this displayed terminal hold.

F.3 Quantum musical-transition prediction

Definition F.2 (Complexity conventions). Fix the standard finite universal gate set $\mathcal{G} = \{H, T, C_{\mathrm{NOT}}\}$, where

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad T = \mathrm{diag}(1, e^{\imath\pi/4}), \qquad C_{\mathrm{NOT}}|c, r\rangle = |c, r \oplus c\rangle.$$

A promise problem is a pair $\mathcal{L} = (\mathcal{L}_{\text{yes}}, \mathcal{L}_{\text{no}})$ of disjoint languages. We write PromiseBQP for the class of promise problems decided by a polynomial-time uniform family of polynomial-size quantum circuits over $\mathcal{G}$. There are polynomial-time computable rational thresholds $0 \le \alpha(\ell) < \beta(\ell) \le 1$, separated by an inverse polynomial in the classical input length ℓ, such that acceptance probability is at least $\beta(\ell)$ on $\mathcal{L}_{\text{yes}}$ and at most $\alpha(\ell)$ on $\mathcal{L}_{\text{no}}$. Repetition on fresh registers and reversible comparison of the observed frequency with the thresholds' midpoint give the usual bounded-error definition with polynomial overhead [46, Sec. IV.2].

Remark F.3 (History presentation). For the feedback architecture in the main text, fix a finite audible alphabet A_{snd} and a literal polynomial-time injective encoding enc of circuit descriptions into finite symbol strings. Choose this encoding so that the symbol string is at least as long as the circuit's bit description. A history $m = (a_1, \dots, a_t) \in \mathsf{A}^*_{\text{snd}}$ consists only of symbols that have already been played. The special prelude

$$m_Q \coloneqq \operatorname{enc}(\langle Q \rangle) \tag{193}$$

is decoded by the score compiler before the next passage, giving $U(m_Q) = Q$. Conversely, the decoded circuit supplies the next transition prediction. Encoding and decoding are classical polynomial-time operations, so this history presentation and the explicit circuit presentation below define polynomial-time equivalent presentations when the thresholds a, b are passed as additional input. Exact identity gate pairs can pad the decoded circuit, or its encoded history, to ensure that the target input length is at least the source length, preserving the inverse-polynomial gap condition. Symbols produced by a passage are appended only for a later compiler invocation; no same-run quantum feedback is assumed.

We now formulate the prediction problem and prove its completeness locally. Fix two distinct terminal transition labels $\mathsf{t}_0, \mathsf{t}_1$; these are classical score outputs, not circuit gates or time indices. An instance supplies a classical description $\langle Q \rangle$ of an n-qubit circuit over $\mathcal{G}$, initialized in $|0^n\rangle$, and rational thresholds a, b. We use an explicit circuit encoding whose length polynomially bounds both the number of qubits and the number of gates. Measuring its distinguished musical output qubit M gives $z \in \{0,1\}$ and selects t_z; the remaining qubits form an unobserved register J, with outcome $u \in \{0,1\}^{n-1}$. In the computational atomic context, set

$$\ell_{\text{tr}}(z, u) \coloneqq \mathsf{t}_z. \tag{194}$$

The probability of the designated musical transition t_1 is therefore

$$p_Q \coloneqq \Pr_Q(M = 1) = \sum_{u \in \{0,1\}^{n-1}} [\mathsf{B}_{\mathcal{C}}(Q)]_{(1,u),0^n}. \tag{195}$$

The explicit context subscript records that this is the canonical computational-context Born kernel.

Definition F.4 (Quantum musical-transition prediction (QMTP)). Fix an integer $c \ge 1$. An instance is $\langle Q, a, b \rangle$, with rational thresholds $0 \le a < b \le 1$ satisfying $b - a \ge \ell^{-c}$, where $\ell \coloneqq |\langle Q, a, b \rangle|$ is the total input bit length. The thresholds are part of the input; c is fixed for the problem. Among these instances, the yes- and no-instances are

$$\begin{aligned} \mathrm{QMTP}_{\mathrm{yes}} &:= \{\langle Q, a, b\rangle : p_Q \geq b\}, \\ \mathrm{QMTP}_{\mathrm{no}} &:= \{\langle Q, a, b\rangle : p_Q \leq a\}. \end{aligned} \tag{196}$$

Inputs with $a < p_Q < b$ lie outside the promise. The promise gap is $b - a$.

Theorem F.5 (PromiseBQP-completeness).

For every fixed $c \geq 1$ in Definition F.4, quantum musical-transition prediction (QMTP *) is* PromiseBQP*-complete under deterministic classical polynomial-time many-one reductions.*

Proof. For membership, run Q independently N times, measure M in each run, and compare the observed frequency with $(a+b)/2$. Hoeffding's inequality bounds the error by $\exp[-N(b-a)^2/2]$, so $N = O(\ell^{2c})$ repetitions give bounded error. Reversible counting and rational-threshold comparison implement this procedure as a uniform polynomial-size quantum circuit.

For hardness, let $\mathcal{L} = (\mathcal{L}_{\mathrm{yes}}, \mathcal{L}_{\mathrm{no}}) \in$ PromiseBQP. Its uniform circuit family produces in classical polynomial time an $n = n(|w|)$-qubit verifier $U(w)$, with both qubit and gate counts polynomially bounded in $|w|$, over $\mathcal{G}$, including the computational-basis preparation of w, with acceptance qubit s, recorded outcome $S \in \{0,1\}$, and remaining work register W. Let $\alpha_w < \beta_w$ be its polynomial-time computable rational thresholds, with $\Delta_w := \beta_w - \alpha_w \geq 1/r(|w|)$ for some fixed polynomial r. Then

$$w \in \mathcal{L}_{\mathrm{yes}} \Longrightarrow \Pr(S=1) \geq \beta_w, \qquad w \in \mathcal{L}_{\mathrm{no}} \Longrightarrow \Pr(S=1) \leq \alpha_w. \tag{197}$$

Add a fresh qubit M in $|0\rangle$ and define

$$Q(w) := C_{\mathrm{NOT}}^{s \to M} \circ (U(w) \otimes \mathbb{1}_M). \tag{198}$$

Here $Q(w)$ has $n+1$ qubits, and the unobserved register above is $\mathsf{J} = (\mathsf{W}, s)$. Writing the verifier state before the final CNOT as

$$U(w)|0^n\rangle = |\phi_{w,0}\rangle_{\mathsf{W}}|0\rangle_s + |\phi_{w,1}\rangle_{\mathsf{W}}|1\rangle_s, \tag{199}$$

with unnormalized branch vectors, gives $p_{Q(w)} = \|\phi_{w,1}\|^2 = \Pr_{U(w)}(S=1)$. Set $a(w) = \alpha_w$ and $b(w) = \beta_w$. Pad the explicit circuit description with $H \circ H = \mathbb{1}$ gate pairs if needed so that $\ell(w) := |\langle Q(w), a(w), b(w)\rangle| \geq \lceil \Delta_w^{-1} \rceil$. This preserves the unitary and its acceptance probability exactly and uses only polynomially many gates, since $\Delta_w^{-1} \leq r(|w|)$. For $c \geq 1$, it gives $\ell(w)^{-c} \leq \ell(w)^{-1} \leq \Delta_w$. Thus $w \mapsto \langle Q(w), a(w), b(w)\rangle$ is a classical polynomial-time reduction satisfying the QMTP promise. Composing it with $Q(w) \mapsto m_{Q(w)} = \mathrm{enc}(\langle Q(w)\rangle)$, retaining $a(w), b(w)$ and padding if needed, proves the same hardness statement when QMTP is presented as an already played musical history. □

Thus, in the standard promise-class sense, deciding a designated transition in this circuit-universal quantum score model is PromiseBQP-complete [47]. This neither makes every fixed transition hard, nor proves that the restricted Born-kernel scores here are universal or that deciding cue stability is PromiseBQP-hard.

F.4 A verifier-controlled stepwise check with an unconditional endpoint

In Figure 8, W is the verifier work register, D is the data register, and M_{step} and M_{end} retain the intermediate and endpoint outcomes Y_{step} and $Y_{\text{end}} := Y$. The fixed history argument m is suppressed in the gate labels.

The verifier output can also choose a measurement schedule. Let m be a fixed already played history, and let its score compiler produce a selector circuit $U(m)$ with acceptance qubit s and $p(m) := \Pr_{U(m)}(S = 1)$. The same compiler selects a pair $U_1(m), U_2(m)$ of k-qubit data notes. Let $D = (q_0, \dots, q_{k-1})$ be a k-qubit data register initialized in $|x\rangle$, with $x \in X = \{0, \dots, 2^k - 1\}$ labeling a computational-context outcome, and let the two data notes act on D in that order. Initialize the meter registers M_{step} and M_{end} in $|0^k\rangle$.

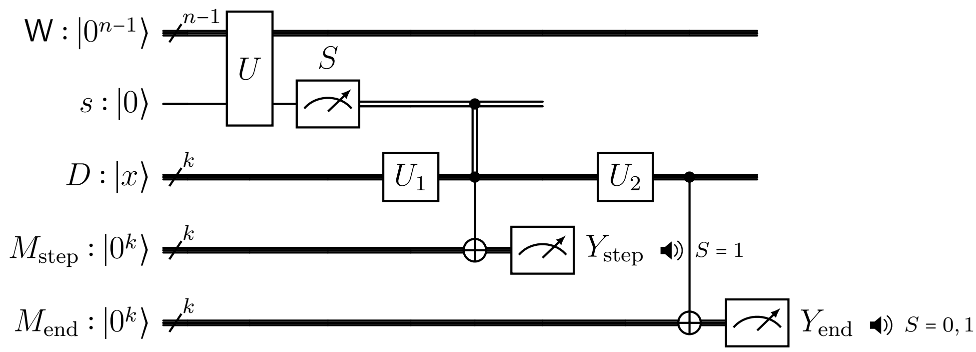

Figure 8: **Verifier-controlled checking schedule.** Reading S before the data passage classically enables the step record bank only for $S = 1$; its retained outcome Y_{step} supplies the optional intermediate sound. The endpoint bank records and sounds Y_{end} on both branches. The intermediate outcome is not used as feed-forward within the passage, although its measurement backaction on D is retained. Bundled wires denote multiqubit registers.

Write $\mathrm{CCX}_{c_1, c_2 \to r}$ for the Toffoli gate

$$|c_1, c_2, r\rangle \longmapsto |c_1, c_2, r \oplus (c_1 c_2)\rangle.$$

It has an exact constant-size decomposition over $\mathcal{G}$ $(T^\dagger = T^7)$. Apply

$$\begin{aligned} C^{(s)}_{\text{step}} &:= \prod_{j=0}^{k-1} \mathrm{CCX}_{s, q_j \to (M_{\text{step}})_j}, \\ C_{\text{end}} &:= \prod_{j=0}^{k-1} C_{\text{NOT}}^{q_j \to (M_{\text{end}})_j}. \end{aligned} \tag{200}$$

Place $C^{(s)}_{\text{step}}$ after $U_1(m)$ and the unconditional C_{end} after $U_2(m)$. If S is the recorded value of s, $X = x$ is the data entrance, and $Y = y$ is the endpoint record, then summing over the step record gives

$$\begin{aligned} K_0(m) &= \mathsf{B}_{\mathcal{C}}(U_2(m) \circ U_1(m)), \\ K_1(m) &= \mathsf{B}_{\mathcal{C}}(U_2(m)) \circ \mathsf{B}_{\mathcal{C}}(U_1(m)), \end{aligned} \tag{201}$$

$$\begin{aligned}\Pr(S=0,Y=y\mid X=x,m)&=(1-p(m))(K_0(m))_{yx},\\ \Pr(S=1,Y=y\mid X=x,m)&=p(m)(K_1(m))_{yx}.\end{aligned}\tag{202}$$

Thus the endpoint check is present on both branches, whereas the stepwise check is present only on the accepting branch. The step record may be read and forgotten at the boundary, or its measurement may be deferred to the end: because no later gate acts on that record, both implementations give the same endpoint statistics. Figure 8 shows the equivalent early-readout implementation for live playback.

For live playback, measure s first and classically enable the step CNOT bank; this implements the same conditional endpoint law. Enable the step speaker only for $S=1$ and the endpoint speaker for both values. The emitted symbols are appended to m only after this passage; marginalizing the step record therefore leaves Eq. (202) unchanged within the current run.

Corollary F.6 (Endpoint-only conditional-check prediction). *Specialize Fig. 8 to $k=1$, initialize D in $|0\rangle$, and take $U_1(m)=U_2(m)=H$. Fix an integer $c\geq 1$. An instance is $\langle m,a,b\rangle$, where m uses the circuit-universal history presentation of Remark F.3 and its compiler produces a verifier circuit $U(m)$ over $\mathcal{G}$ with designated output qubit s. The rational input thresholds satisfy $0\leq a<b\leq 1$ and $b-a\geq \ell^{-c}$, where $\ell := |\langle m,a,b\rangle|$ is the total input length. The task is to decide whether*

$$\Pr(Y=1)\geq\frac{b}{2}\qquad or \qquad \Pr(Y=1)\leq\frac{a}{2},\tag{203}$$

It is promised that one alternative holds. This problem is PromiseBQP*-complete under deterministic classical polynomial-time many-one reductions.*

Proof. On the branch $S=0$, no step record is created and $H\circ H|0\rangle=|0\rangle$. On the branch $S=1$, the step record dephases $H|0\rangle=|+\rangle$ in the computational context; the probe is then entangled with the record, so its reduced state is maximally mixed. The second Hadamard leaves that reduced state maximally mixed. Hence

$$\Pr(Y=1\mid S=0)=0,\qquad \Pr(Y=1\mid S=1)=\frac{1}{2},\qquad \Pr(Y=1)=\frac{1}{2}p(m).\tag{204}$$

The gadget sends the verifier thresholds a,b to $a/2,b/2$ and halves their separation, leaving an inverse-polynomial gap. For hardness, encode the verifier and pass its thresholds exactly as in the proof of Theorem F.5, padding with identity gate pairs when needed for the total history-instance length. Equation (204) then gives the required endpoint promise.

For membership, sample N independent runs and compare the endpoint frequency with $(a+b)/4$. The error is at most $\exp[-N(b-a)^2/8]$, so $N=O(\ell^{2c})$ gives bounded error. □

If the schedule bit S is retained, this H–H specialization also gives the explicit information value

$$I(S;Y)=H_2\left(1-\frac{p(m)}{2},\frac{p(m)}{2}\right)-p(m)\quad\text{bits per run},\tag{205}$$

which is the biased-schedule specialization of Eq. (128). This identity links the two constructions operationally; the completeness proof itself is the circuit reduction above.

F.5 Exact sampling of musical outputs

Remark F.7 (Complexity-class terminology). BPP and BQP are the bounded-error classical randomized and quantum polynomial-time language classes. An IQP circuit is a polynomial-size circuit of mutually commuting gates diagonal in the Pauli- X basis, followed by computational-basis measurement. The polynomial hierarchy PH is the union of its alternating quantifier levels; its third level is $\Sigma_3^{\mathrm{P}} \cup \Pi_3^{\mathrm{P}}$. These standard notions are recalled only to state the sampling consequence below.

Proposition F.8 (Exact musical-output sampling). *Fix a polynomial-time injective encoding of full computational-basis output strings into polynomial-length musical strings, with a polynomial-time inverse. If a classical randomized polynomial-time algorithm, given the circuit description, samples these musical outputs exactly for every circuit in the uniform finite-gate IQP families of Bremner, Jozsa, and Shepherd [48], then the polynomial hierarchy collapses to its third level.*

Proof. Applying the inverse encoding to each sample gives an exact classical sampler for the full IQP output distribution. The circuit families needed in [48] use only a finite set of one- and two-qubit gates. Their exact-sampling consequence therefore applies directly. □

This concerns sampling full output strings. By contrast, an efficient classical randomized decider for the QMTP promise would imply BQP $\subseteq$ BPP; its decision-completeness theorem alone does not establish a collapse of the polynomial hierarchy.

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